

Semantic Time Geography: Bringing Semantic Contexts to Notation System of Time Geography

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Abstract

Time geography is a powerful framework. It provides an elegant notation system with a series of well-defined space-time entities and relationships for representing human activity-travel behaviors under various space-time constraints. The classical time-geographical framework focuses on only the geometric aspect of activity-travel behaviors in the physical space but ignores semantic contexts about activity-travel behaviors in the hybrid cyber-physical space (e.g., activity goals, transport modes, and personal emotions and satisfactions). This study proposes a novel framework of semantic time geography by introducing a series of semantically enriched time-geographical entities (e.g., semantic space-time paths, prisms, lifelines, and projects) and relationships (e.g., semantic space-time intersection relationships) in the hybrid cyber-physical space. The proposed framework is implemented by using the relational database and integrated with an existing GIS platform. A new semantic accessibility measure is introduced to illustrate the applicability of the proposed framework. Experimental results demonstrate the capabilities of the introduced entities and relationships for representing, visualizing, querying, and analyzing both space-time and semantic dimensions of human activity-travel behaviors in the hybrid cyber-physical space.

Keywords: time geography, spatiotemporal data model, semantic time geography, behavioral geocomputation

1. Introduction

Time geography is a powerful framework for understanding individuals' activity-travel behaviors under various space-time constraints (Miller, 2005a). It provides an elegant notation system with a series of well-defined space-time entities and relationships to represent activity-travel behaviors. This framework has been applied in broad applications, e.g., behavioral data mining, accessibility studies, travel demand modeling, social equity, social segregation, etc. (Farber et al., 2013).

The classical time-geographical framework was initially proposed by Torsten Hägerstrand in the 1960s. Hägerstrand mainly focused on the understanding of individual activities through the lens of constraints (Hägerstrand, 1970). He defined a set of conceptual space-time entities to represent activity-travel behaviors, such as space-time paths, prisms, and lifelines. The space-time path, also named trajectory, represents individual experienced movements and activities in space and time dimensions. The space-time prism delimits all potential space-time regions to schedule a flexible activity between two fixed activities. Hägerstrand further defined a set of space-time relationships for space-time entities, e.g., intersection and bundling. However, time-geographical studies, to some extent, faded into the background in the decades after its introduction, due to the lack of individual-

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level data and computational tools (Long and Nelson, 2013).

With the development of Geographical Information Science (GIS), we have witnessed a resurgence of time-geographical studies in the last three decades. The analytical foundation and measurement theory was established for time-geographical entities (Miller, 2005a). Much research effort was devoted to developing GIS-based tools to represent, compute, and analyze time-geographical entities and relationships (Miller, 1991; Kwan and Weber, 2003; Miller and Bridwell, 2009; Chen et al., 2013; Liao, 2021). The space-time cube model and corresponding visual analytics techniques were developed to visualize and explore time-geographical phenomena (Nakaya, 2013). In recent years, there has been a wider resurgence of time-geographical studies fueled by emerging spatiotemporal big data, including taxi trajectories, smart card data, and mobile phone tracking data (Luo et al., 2025). Such big data have provided golden opportunities for time-geographical studies, which build on individual-level data. Researchers developed effective and efficient spatiotemporal data models and index structures to store, manage, query, and process a large number of time-geographical entities (Chen et al. 2016). Data mining techniques were further developed for knowledge discovery from time-geographical entities and relationships (Yuan et al., 2018; Miller et al., 2019; Luo et al., 2025).

The classical notation system of time geography effectively describes space-time dimensions of activity-travel behaviors. However, it has been critiqued as overly materialistic and geometric, neglecting the vital semantic contexts, such as activity goals, subjective perceptions and emotions, and environmental conditions (Ellegård, 1999; 2018; Shaw and Yu, 2009; McQuoid and Dijst, 2012). Incorporating these semantic contexts can greatly enhance the analytical capabilities of time geography for understanding subjective and/or cognitive dimensions of activity-travel behaviors and complicated interactions with surrounding environments. Several researchers have tried to incorporate semantic contexts in time-geographical studies. For example, Kwan (2008) conducted a small case study by collecting individual emotions along the space-time path and analyzing the post-September 11 experiences of Muslim women. Along the same line, McQuoid and Dijst (2012) investigated the emotional experiences of low-income single mothers along their space-time paths. Ellegård and Cooper (2004) developed effective techniques to visualize individual daily activity patterns with contextual information. Although previous works have used semantic contexts for specific analyses or visualizations, little attention has been given to developing a generalizable and computable notation system where semantic contexts are formalized as integral components of the data model itself.

The second critical limitation of the classic notation system of time geography is the overlook of online activities in cyberspace. Online activities—such as e-commerce, telecommuting, and online socializing—have become prevalent in our daily lives, bringing about changes in the characteristics of daily activities (Shaw and Yu, 2009). While the fundamental conditions (such as the indivisibility of individuals) have yet to change, space-time constraints of daily activities may be relaxed at several conditions, due to the use of information and communication technologies (ICTs). Several types of daily activities (e.g., shopping and socializing) could be freely organized in either physical or mediated presence. Built on a dichotomous perspective between physical and cyber spaces, many studies investigated the impacts of ICTs on online activities, including substitution, complementarity, modifications, and neutrality (Zhai et al., 2017; Xi et al., 2020). By using an integrated perspective of physical and cyber spaces, recent studies further revealed intertwined characteristics of many online and offline activities in the hybrid cyber-physical space, such as online shopping and socializing activities conducted during a commuting trip (Thulin et al., 2020; Shaw, 2023). This hybridization of online and offline activities significantly increases the fragmentation and complexities of daily activities (Li et al. 2024). However, few studies have been conducted to extend the classical notation system of time geography for representing semantically enriched activity-travel behaviors in hybrid cyber-physical spaces.

In view of these limitations, this study aims to propose a novel notation system of time geography by introducing a series of semantically enriched entities and relationships to fully represent activity-travel behaviors in hybrid cyber-physical spaces. To distinguish it from the classical notation system of time geography, the proposed notation system is hereafter referred to as the framework of “semantic time geography”; and the introduced entities and relationships are referred to as semantic time-geographical entities and relationships, e.g., semantic space-time paths, semantic space-time prisms, and semantic space-time bundling.

In the literature, much effort has been devoted to developing effective data models for representing semantic space-time paths, also called semantic trajectories. One of the earliest models is the stop-move model (Spaccapietra et al., 2008). It separates a raw trajectory (with only geometric data) into a sequence of stop and move segments, in which the object is stationary or moving, respectively. The semantic trajectory is constructed through a semantic enrichment process by attaching semantic contexts to stop and move segments. However, this stop-move model lacks generality due to its ignorance of other relevant semantic contexts (Fileto et al., 2013). Yan et al. (2013) and Bogorny et al. (2014) proposed a generalized model built on the concept of episodes, which are trajectory segments with homogeneous semantic contexts (e.g., travel speeds, accelerations, and environment factors). Multiple semantic dimensions (e.g., activity goal) are also introduced in the episode model for representing activity-travel behaviors. Following this episode model, various methods have been developed to implement the semantic enrichment process by linking to additional background data, such as open street maps, points of interest (POIs), and social media data (Yan et al., 2013). Querying methods were also developed for performing searches on semantic trajectories by using ontologies (Fileto et al., 2015) and resource description framework (Izquierdo et al., 2021). Further, various data mining methods (e.g., clustering, classification, and outlier detection of semantic trajectories) have been developed by explicitly considering the semantic contexts of individual activities (Renso et al., 2021). However, current semantic trajectory techniques focused on only semantic space-time paths but ignored other time-geographical entities and relationships. Further, current semantic trajectory techniques ignore online activities, which are commonly encountered in today’s Internet society (Thulin et al., 2020; Shaw, 2023; Li et al., 2024).

In this study, we propose a novel framework of semantic time geography, in which a series of semantically enriched entities and relationships are introduced to fully represent activity-travel behaviors in hybrid cyber-physical spaces. The proposed framework provides a powerful tool for understanding not only space-time dimensions but also subjective and/or cognitive dimensions of activity-travel behaviors and complicated interactions with surrounding environments. This study contributes to existing time-geographical literature in the following aspects.

Firstly, the proposed framework explicitly represents both geometric and semantic aspects of activity-travel behaviors in the physical space. It provides rigorous definitions for a new set of semantic space-time entities in the physical space, including semantic space-time paths, stations, prisms, and lifelines. These semantic space-time entities are flexible to record any type of semantic contexts at various levels of space-time granularities. Further, it defines several new semantic space-time relationships between entities by considering not only the coexistence of entities in space-time dimensions but also the match of activity semantics. Therefore, the proposed framework extends the existing notation system of time-geography (Miller, 2005a) by explicitly incorporating rich semantic contexts on activity-travel behaviors in the physical space.

Secondly, the proposed framework provides a series of entities to explicitly represent activity-travel behaviors in the hybrid cyber-physical space, including semantic website-time paths, semantic hybrid activity patterns, and hybrid projects. The semantic website-time path entity is to represent a sequence

of online activities performed by an individual. The semantic hybrid activity pattern entity is to represent a sequence of online and offline activities in a hybrid cyber-physical space. The hybrid project entity represents online and/or offline activities performed by one or more persons for a common goal. Therefore, the proposed framework provides valuable methodologies for supporting time-geographical studies in hybrid cyber-physical spaces.

Thirdly, the proposed framework is implemented by using a relational database and space-time cube model. The logic model of the relational database is designed to represent the proposed semantic time-geographical entities and relationships. Four types of queries are implemented to perform queries on semantic time-geographical entities. The proposed framework is further integrated with GIS platforms for visual analysis. To demonstrate the applicability of the proposed framework, a new semantic accessibility analysis method is introduced by using semantic space-time interaction relationships between prisms and stations. Results highlight the capabilities of the proposed framework for representing, visualizing, querying, and analyzing both geometric and semantic perspectives of human activity-travel behaviors in the hybrid cyber-physical space.

2. Framework of semantic time geography

In this section, we describe the proposed framework of semantic time geography by explicitly incorporating rich semantic information into the framework of classical time geography. As shown in Figure 1, the proposed framework introduces a series of entities and relationships in three spaces, including physical space, cyberspace, and social space. For the physical space, we define two categories of entities: transportation systems and land-use systems. Transportation systems represent infrastructures of various transport modes to facilitate human movements from one location to another in the physical space. Land-use systems refer to physical venues for performing individual daily activities, such as working, shopping, sleeping, etc. Analogous to the physical space, we define two categories of entities in cyberspace: cyberspace infrastructures and cyberspace apps. Cyberspace infrastructures refer to physical devices providing high-speed internet access or data transmissions, such as Wifi access points. These cyber infrastructures also have spatial locations in the physical space. Cyberspace apps represent numerous online applications for performing individual online activities, e.g., online shopping and online social activities. Typical cyberspace apps include YouTube, Google Maps, Facebook, WeChat, Amazon, UberEats, etc. All these entities defined in both physical space and cyberspace together provide a hybrid cyber-physical world within which individuals can engage in various offline and/or online activities.

For the social space, we define three types of semantic time-geographical entities: individuals, social organizations, and individual activity-travel behaviors. Individual is a basic entity in the classical time-geographical framework. This entity is enriched with semantic information with respect to individual characteristics. Social organization entities are introduced to represent social units encapsulating certain social relationships among their members, such as households and companies. Individual activity-travel behaviors include a series of semantic time-geographical entities representing individual offline and/or online activities in the hybrid cyber-physical space. We define two types of semantic time-geographical relationships, semantic social relationships, and semantic space-time relationships (including space-time relationships and activity semantic relationships), to describe relationships among entities in the social space.

In addition to entities defined in three spaces, we further define two types of entities as external factors: society rules and environmental conditions. Society rules represent social institutions and policies that can influence our daily routines, such as teleworking policies. These society rules explicitly implement the concept of authority constraints in the time geography field. Environmental conditions represent environmental constraints affecting our daily activities, such as weather conditions and air pollution.

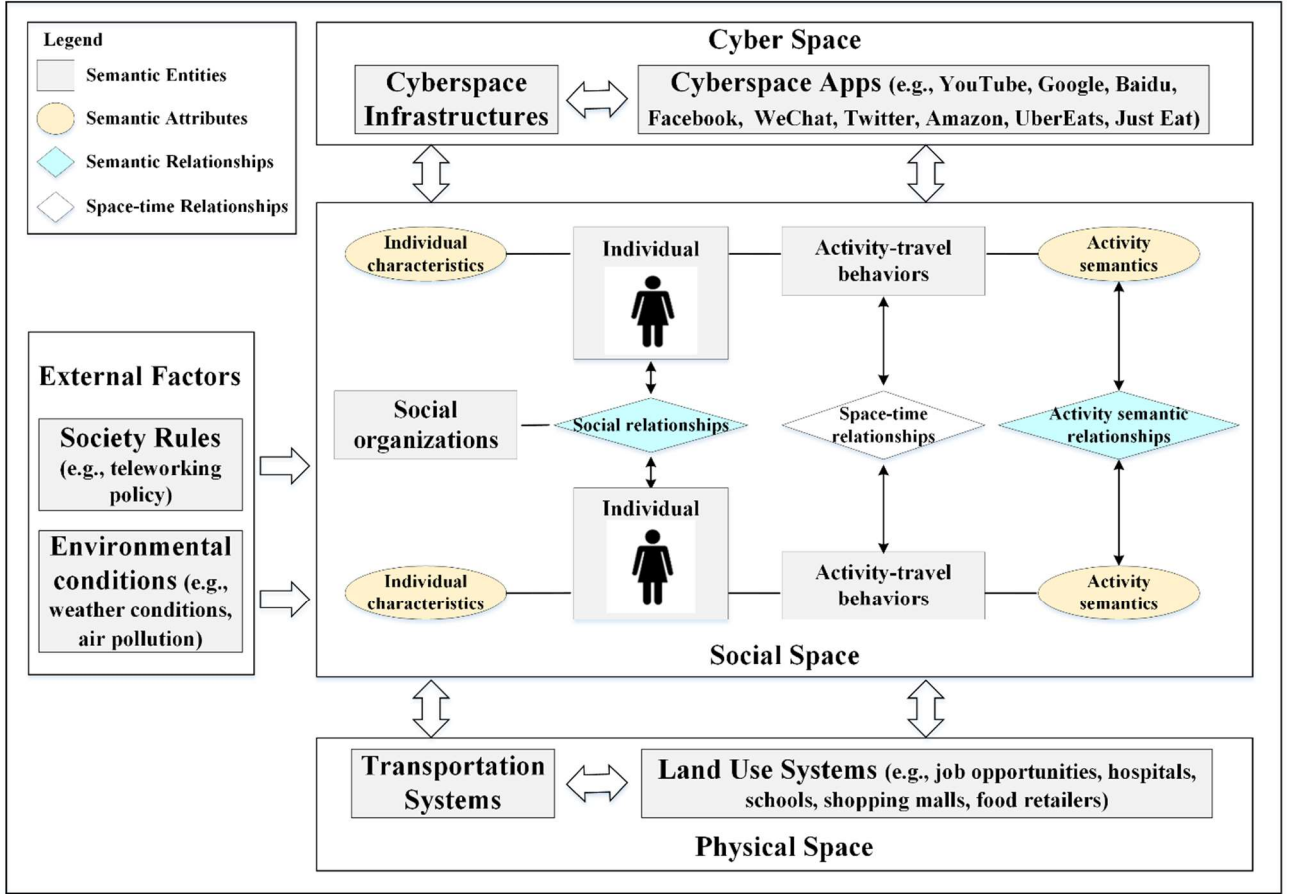


Figure 1. Framework of proposed semantic time geography.

In the proposed framework, semantic contexts are explicitly incorporated to formulate semantically enriched activity-travel behaviors in the hybrid cyber-physical space. Typical semantic contexts include individual characteristics and subjective perceptions, activity semantics, travel status, rich facility information, etc. To simplify the representation, we provide essential definitions of all semantic time-geographical entities and relationships defined in the social space in the following sections but give definitions of other entities in the supplementary sections.

3. Semantic time-geographical entities defined in the social space

This section describes semantic time-geographical entities defined in the social space. As shown in Figure 2, the introduced entities can be classified into three categories. The first category, shown in yellow color, has three social organization entities, including Individuals (SemIndividual), households (SemHousehold), and companies (SemCompany). The second category, shown in purple color, consists of four entities representing activity-travel behaviors in the physical space, including semantic space-time paths (SemPath), prisms (SemPrism), lifelines (SemLifeLine), and stations (SemStation). In the figure, PhyEntity is the generic superclass of these four entities in the physical space. The last category, shown in light blue color, consists of three entities representing activity-travel behaviors in the hybrid cyber-physical space, including the semantic website-time paths (VirPath), semantic hybrid activity patterns (HybActivityPattern), and hybrid projects (HybProject). The detailed descriptions of entities in three categories are given in the following sections. A summary of these semantic entities is given in Supplementary Section S3 with a comparison to entities defined in the classical notation system (Miller, 2005a).

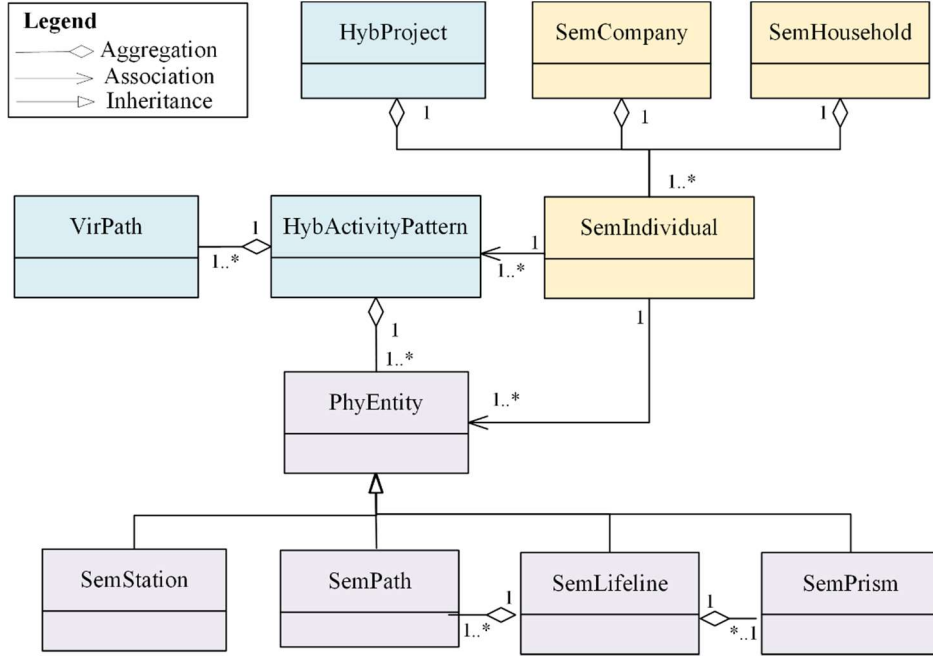


Figure 2. Overview of semantic time-geographical entities defined in the social space.

3.1. Individuals and social organizations

This section describes three entities of social organizations shown in Figure 2 in yellow. The individual entity, denoted by $SemIndividual^q$, represents an indivisible object that performs activity-travel behaviors in the cyber-physical space. An $SemIndividual^q$ object can have semantic information in terms of socio-economic attributes, residential location, and workplace. It can have multiple social relationships, such as household memberships and colleague relationships, and can have activity-travel behaviors. It can be expressed as:

$$SemIndividual^q = \{SocialRelSet^q, DailyActivitySet^q, sm_SemIndividual^q\} \quad (1)$$

where $SocialRelSet^q$ is the set for recording individual social relationships (described in Section 4.1); $DailyActivitySet^q$ is the set for recording individual daily activities for multiple days (described in Section 3.3), and $sm_SemIndividual^q$ is the set recording individual semantic information. We set $sm_SemIndividual^q = \{Name, FamilyRole, Home, Occupation, Workplace, hasDrivingLicence, OtherContextSet\}$. The *FamilyRole* attribute represents the role of the person within his/her household, such as male head, female head, and child. The *Home* and *Workplace* attributes are $SemPlace_j$ entities representing his/her residential location and workplace (or school for a child), respectively. The *hasDrivingLicence* attribute is a binary variable indicating whether the person has a driving license. The *OtherContextSet* provides a flexible way to include various semantic contexts from specific applications.

The household entity, denoted by $SemHousehold^h$, consists of multiple household members and has semantic information in terms of household attributes. It can be expressed as

$$SemHousehold^h = \{MemberSet^h, sm_semHousehold^h\} \quad (2)$$

where $MemberSet^h$ is the set for recording members in the household; and $sm_semHousehold^h$ is the set for recording semantic information of the household. We set $sm_semHousehold^h = \{Home, MemNum, HouIncome, CarNum, OtherContextSet\}$. The *MemNum* attribute represents the number of household members; the *HouIncome* attribute represents the level of household income; and the *CarNum* attribute represents the number of private cars owned by the household. Similar to the household entity, the company entity, denoted by $SemCompany^c$, consists of multiple company members and has semantic information in terms of company attributes. The detailed definition of this entity is given in Supplementary Section S4.

3.2. Activity-travel behaviors in the physical space

This section describes semantic time-geographical entities representing individual activity-travel behaviors in the physical space. Figure 3 gives the detailed entities for representing four semantic time-geographical entities, including semantic space-time path (in yellow color), semantic space-time station (in grey color), semantic space-time prism (in light blue color), and semantic space-time lifeline (in purple color). The descriptions of these entities are given in the following sections.

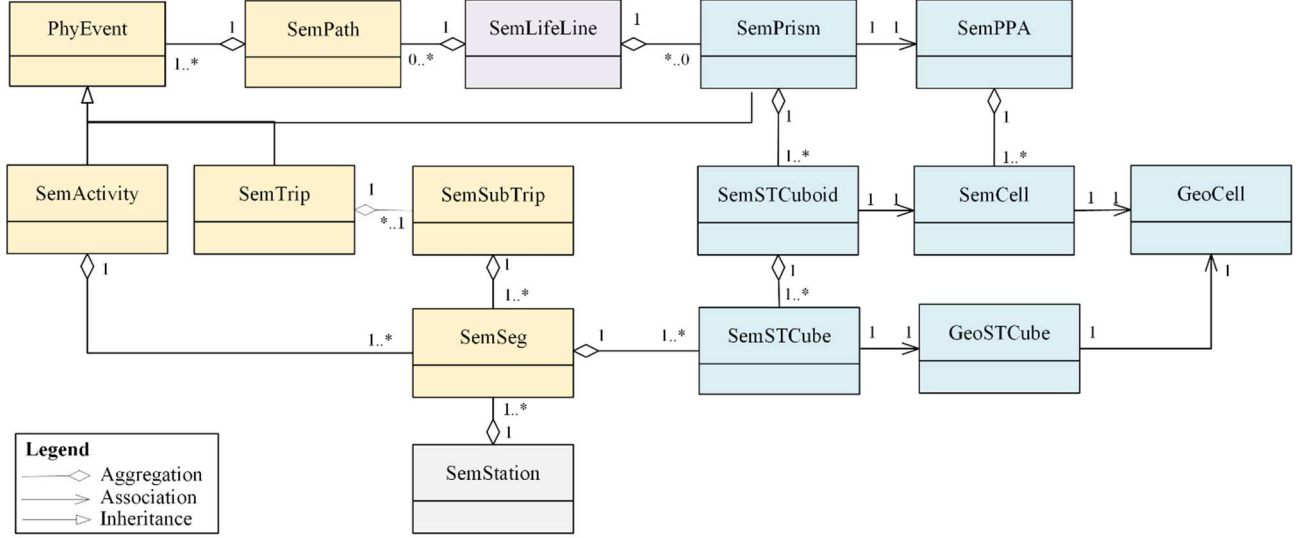


Figure 3. Detailed entities representing activity-travel behaviors in the physical space.

3.2.1. Semantic space-time path entity

This section describes the semantic space-time path entity (denoted by $SemPath_k^q$). It extends the classical space-time path concept by incorporating semantic information. During activities and/or movements in the physical space, the individual could have multiple semantic information (such as travel speeds, emotions, transport mode, etc.) at different space-time granularities. To maintain such semantic information on an $SemPath_k^q$ object, we introduce an additional four semantic space-time entities. Following the episode model (Spaccapietra et al. 2008; Fileto et al. 2013), we first define the semantic line segment, denoted by $SemSeg_i^q$, represents a movement with homogeneous status in multiple contexts. It can be expressed as a tuple:

$$SemSeg_i^q = \{GeoSeg_{ij}, sm_SemSeg_i^q\} \quad (3)$$

where $GeoSeg_{ij} = \{c_i, c_j\}$ is the classical line segment from space-time points c_i to c_j , which could be a straight line, a curve, or a portion of a network link. $sm_SemSeg_i^q$ is the set for recording semantic information at the segment. We set $sm_SemSeg_i^q = \{StartTime, EndTime, Duration, Speed, OtherContextSet\}$. The $StartTime$ ($EndTime$) attribute represents the start time (end time) of the activity, and the $Duration$ attribute represents the duration of the activity. The $Speed$ attribute represents the constant travel speed of the line segment.

Following the literature on activity-based travel demand modeling (Fu et al., 2022), we then introduce two semantic space-time entities: $SemActivity$ and $SemTrip$. The $SemActivity$ entity represents the activity carried out at a place, including the waiting time before or after the activity. The i^{th} activity made by individual q , denoted by $SemActivity_i^q$, can be expressed as a tuple:

$$SemActivity_i^q = \{SemSegSet_i^q, sm_SemActivity_i^q\} \quad (4)$$

where $SemSegSet_i^q$ is a set of consecutive line segments as $\{SemSeg_j^q, \dots, SemSeg_k^q\}$; and $sm_SemActivity_i^q$ is the set for recording semantic information of the activity. We set

$sm_SemActivity_i^q = \{ ActivityGoal, SemPlace, StartTime, EndTime, Duration, OtherContextSet \}$. The *ActivityGoal* attribute represents the goal of an activity, such as working, shopping, sporting, etc. The *SemPlace* attribute represents the physical environment where the activity was conducted, such as home and workplace, and its definition is given in Supplementary Section S1.

A *SemTrip* is defined as a trip from an origin to an activity destination. In real applications, a trip can be made by multiple transport modes, e.g., walking, bus, subway, and taxi. Accordingly, we define a *subtrip* entity representing a part of the trip using a specific transport mode. The i^{th} trip made by individual q , denoted by $SemTrip_i^q$, is expressed as a tuple:

$$SemTrip_i^q = \{ SubTripSet_i^q, sm_SemTrip_i^q \} \quad (5)$$

where $SubTripSet_i^q$ is a set of consecutive subtrip entities, and $sm_SemTrip_i^q$ is the set for recording semantic information at the trip level. We set $sm_SemTrip_i^q = \{ StartTime, EndTime, Duration, Origin, Destination, isMultiMode, isCommuTrip, OtherContextSet \}$. The *isMultiMode* attribute is a binary indicator representing whether the trip consists of multiple transport modes, and the *isCommuTrip* attribute is also a binary indicator representing whether it is a commuting trip between home and workplace.

The j^{th} subtrip of $SemTrip_i^q$, denoted by $SemSubTrip_j^i$, is expressed as a tuple:

$$SemSubTrip_j^i = \{ SemSegSet_j^i, sm_SemSubTrip_j^i \} \quad (6)$$

where $SemSegSet_j^i$ is the set of consecutive line segments; and $sm_SemSubTrip_j^i$ is the set for recording semantic information at the subtrip level. We set $sm_SemSubTrip_j^i = \{ StartLoc, EndLoc, StartTime, EndTime, Duration, TransportMode, OtherContextSet \}$. The *StartLoc* and *EndLoc* attributes represent the from and to locations, respectively. It should be noted that subtrips of the same trip should be consecutive. In other words, *EndLoc* and *EndTime* of i^{th} subtrip should be equal to *StartLoc* and *StartTime* of the $i+1^{th}$ subtrip, respectively.

We then define the concept of semantic space-time path (i.e., $SemPath_k^q$). It is defined as an activity pattern performed by an individual q during a particular day k and can be expressed as a tuple:

$$SemPath_k^q = \{ SemEventSet_k^q, sm_SemPath_k^q \} \quad (7)$$

where $SemEventSet_k^q$ is a set of consecutive activities and/or trips. We introduce a generic super-type, *PhyEntity*, for both *SemActivity* and *SemTrip*. Its semantic information set is defined as $sm_SemPath_k^q = \{ TripNum, TotalTravelTime, ActivityNum, TotalOutHomeTime, StartTime, EndTime, Duration, OtherContextSet \}$, where *TripNum* represents the total number of trips; *ActivityNum* represents the total number of activities performed out of home; and *TotalOutHomeTime* represents the total time spent out of home.

Figure 4 illustrates a semantic space-time path $SemPath_k^{Lisa}$ representing the activity pattern performed by Lisa during a typical working day. As shown, Lisa performed four activities outside of the home, including working, having lunch, working, and shopping. She made five trips, and two of them were multi-mode trips. Accordingly, we have $SemEventSet_k^{Lisa} = (SemTrip_1^{Lisa}, SemActivity_2^{Lisa}, SemTrip_2^{Lisa}, SemActivity_3^{Lisa}, SemTrip_3^{Lisa}, SemActivity_4^{Lisa}, SemTrip_4^{Lisa}, SemActivity_5^{Lisa}, SemTrip_5^{Lisa})$. The set of semantic information at the path level can be recorded as $sm_SemPath_k^{Lisa} = \{ 5, 3 \text{ hour}, 4, 13 \text{ hour}, null \}$. The trip, $SemTrip_1^{Lisa}$, is a multi-mode commuting trip from home to work using four transport modes with respect to walking, bus, subway, and walking modes. Then, we have the set of four subtrips as $SubTripSet_1^{Lisa} = (SemSubTrip_1^1, SemSubTrip_2^1, SemSubTrip_3^1 \text{ and } SemSubTrip_4^1)$. The set of semantic

information for this commuting trip can be recorded as $sm_SemTrip_1^{Lisa} = \{Home_1^{Lisa}, Workplace_1^{Lisa}, 07:00, 08:30, true, null\}$. The set of semantic information for the first subtrip, $SemSubTrip_1^1$, can be recorded by $sm_SemSubTrip_1^1 = \{Home_1^{Lisa}, Busstop_i, 07:00, 07:10, Walking, null\}$. During this subtrip, Lisa walked in a happy mood at a constant speed of 5 km/h. Thus, we have a single semantic line segment in $SemSegSet_1^1 = (SemSeg_1^{Lisa})$. The semantic information of the segment can be recorded as $sm_SemSeg_1^{Lisa} = \{07:00, 07:10, 10\ min, 5\ km/h, happy, cloudy\}$. From this example, it is clear that the introduced semantic space-time path (i.e., $SemPath$) is a powerful extension of the classical space-time path ($GeoPath$) to incorporate semantic information at various space-time granularities, i.e., segment, subtrip, trip, activity, and path levels.

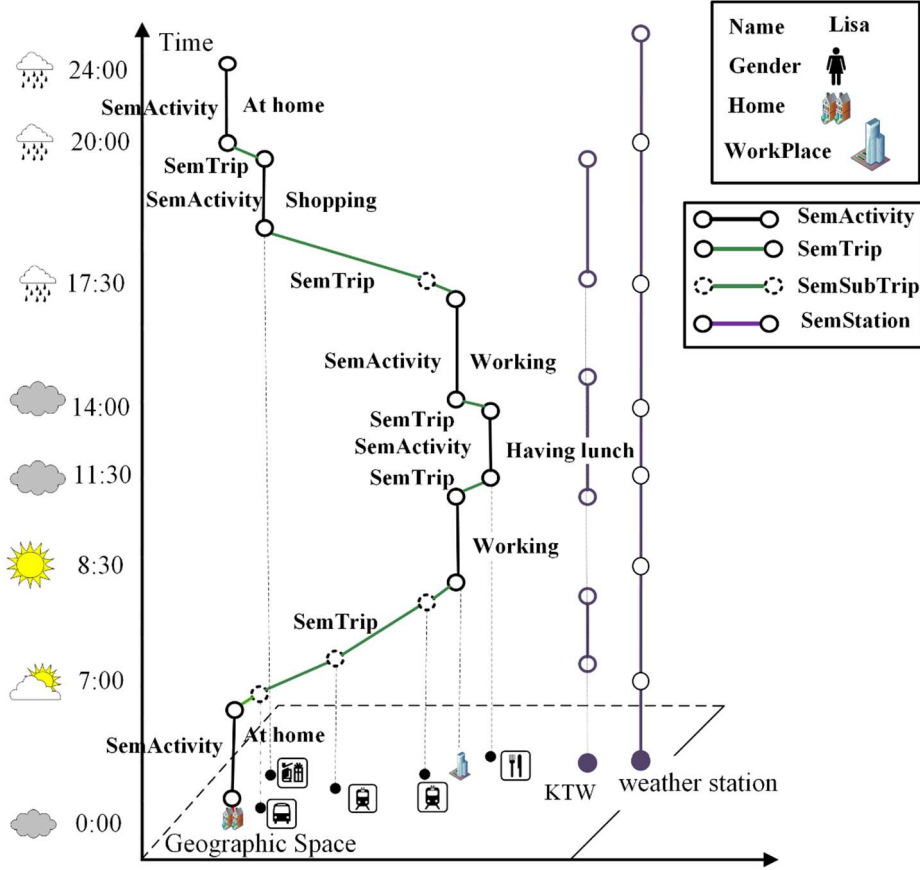


Figure 4. An illustrative example of the semantic space-time path.

3.2.2. Semantic space-time station entity

This section introduces the semantic space-time station entity (denoted by $SemStation^f$), which represents a physical facility (e.g., a shopping center) in the physical space. It is expressed as a tuple:

$$SemStation^f = \{SemSegSet^f, sm_SemStation^f\} \quad (8)$$

where $SemSegSet^f$ is a set of consecutive line segments $\{SemSeg_j^f, \dots, SemSeg_k^f\}$. Each segment $SemSeg_j^f = \{GeoSeg_{ij}, sm_SemSeg_j^f\}$ maintains a vertical line $GeoSeg_{ij}$ located at facility f , and homogenous status in multiple contexts recorded in $SemSeg_k^f$. The contexts of interest are related to time-dependent attributes of the facility, such as its opening status, attractiveness, crowdedness, and waiting times. The static semantic information can be recorded at the set, $sm_SemStation^f = \{SemPOI, ServiceType, OtherContextSet\}$, where $SemPOI$ represents the POI of the facility, and $ServiceType$ represents the service provided by the facility. The illustrative example of $SemStation^f$ is given in Supplementary Section S5.

3.2.3. Semantic space-time prism entity

This section describes the semantic space-time prism entity (denoted by $SemPrism_i^q$) in the physical space. In time geography, the space-time prism is used to delimit all feasible space-time regions for scheduling a flexible activity between two fixed activities at the origin and destination, respectively. Accordingly, we define a semantic space-time prism of individual q (i.e., $SemPrism_i^q$) as a tuple:

$$SemPrism_i^q = \{SemSTCuboidSet_i^q, sm_SemPrism_i^q\} \quad (9)$$

where $sm_SemSTP_i^q$ is the set of semantic information at the prism level. We set $sm_SemSTP_i^q = \{sm_OriSemFixedAct, sm_DestSemFixedAct, sm_SemFlexibleAct, PpaSize, StpVolume, StartTime, EndTime, Duration, OtherContextSet\}$, where $sm_OriSemFixedAct$ and $sm_DestSemFixedAct$ are used to record semantic information of two fixed activities at the origin and destination respectively; $sm_SemFlexibleAct$ is used to record semantic information of the flexible activity; $PpaSize$ represents the size of the potential path area of the prism, and $StpVolume$ represents the volume of the prism. We further define the semantic information of every fixed activity ($sm_OriSemFixedAct$ or $sm_DestSemFixedAct$) as $\{ActivityGoal, SemPlace, TimeInstant, OtherContextSet\}$, where $ActivityGoal$ represents the goal of activity; $SemPlace$ represents the fixed location of the activity; $TimeInstant$ represents the fixed time of the activity. We define the semantic information of $sm_SemFlexibleAct = \{ActivityType, MinActDur, MoneBudget, preferMode, OtherContextSet\}$, where $MinActDur$ represents the minimum duration constraint for performing the flexible activity; $MoneBudget$ represents the monetary budget constraint for the flexible activity; and $preferMode$ represents the transport mode that will be used by individual q for conducting flexible activity.

In Eq. (9), $SemSTCuboidSet_i^q$ is a discrete version of the classical space-time prism enriched with spatial-varying contexts. It consists of a set of semantic space-time cuboids as $\{SemSTCuboid_1^q, \dots, SemSTCuboid_j^q\}$. Each semantic space-time cuboid, $SemSTCuboid_j^q$, represents the potential space-time regions for conducting the flexible activity at a specific grid j and can be expressed as:

$$SemSTCuboid_j^q = \{SemSTCubeSet_j^q, sm_SemSTCuboid_j^q\} \quad (10)$$

where $sm_SemSTCuboid_j^q$ is the set for recording semantic information at the cuboid level. We set it as $\{ArriTripTime, DeparTripTime, MaximumActDur, OtherContextSet\}$, where $ArriTripTime$ is the minimum travel time required for the arrival trip; $DeparTripTime$ is the minimum travel time required for the departure trip; and $MaximumActDur$ is the maximum activity duration for performing the flexible activity.

In Eq. (10), the $SemSTCubeSet_j^q$ element is a set of semantic space-time cubes $\{SemSTCube_1^q, \dots, SemSTCube_i^q\}$ located at grid j . Each semantic space-time cube, $SemSTCube_i^q$, can be expressed as

$$SemSTCube_i^q = \{GeoSTCube_i^q, sm_SemSTCube_i^q\} \quad (11)$$

where $GeoSTCube_i^q$ is the classical space-time cube in space and time dimensions (Nakaya, 2013), and $sm_SemSTCube_i^q$ is the set for recording semantic information at the cube level, such as frequency of visits by the individual, population density, PM_{2.5} indicator, vitality indicator, etc.

Analogous to the classical space-time prism, we can project a $SemPrism_i^q$ object onto the spatial dimensions by removing the time dimension to generate a semantic potential path area, denoted by $SemPPA_i^q$. This $SemPPA_i^q$ entity is an extension of classical PPA by enriching with semantic information. It can be expressed as

$$SemPPA_i^q = \{SemCellSet_i^q, sm_SemPrism_i^q\} \quad (12)$$

where $SemCellSet_i^q$ is the set of semantic spatial cells. Each semantic spatial cell, $SemCell_j^q$, can be expressed as

$$SemCell_j^q = \{GeoCell_j^q, sm_SemSTCuboid_j^q\} \quad (13)$$

where $GeoCell_j^q$ is the classical spatial cell (or raster cell).

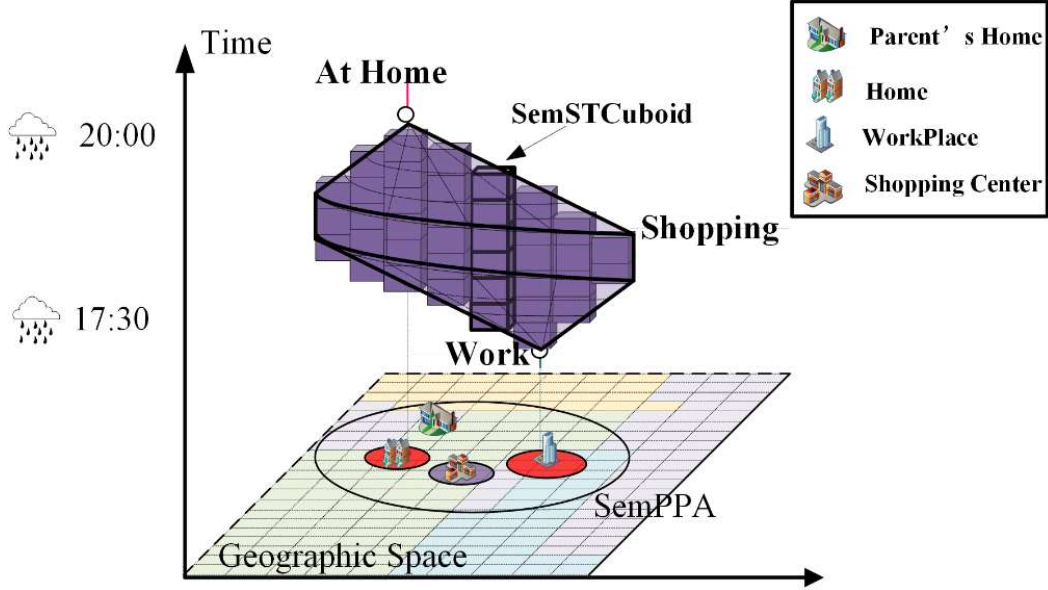


Figure 5. An illustrative example of a semantic space-time prism.

Figure 5 illustrates an example of a semantic space-time prism, $SemPrism_4^{Lisa}$. In this example, Lisa will finish the working activity at her workplace at 17:30, and she is planning to reach home at 20:00 to join a party with her parent. During this period, she would like to schedule a shopping activity, which is flexible in its location, time, and duration. She has a private car and plans to spend at least 2 hours buying a high-quality and expensive gift for her parent. Therefore, the semantic information of a fixed activity at the origin can be recorded as $sm_OriSemFixedAct_3^{Lisa} = \{Working, Workplace_1^{Lisa}, 17:00, null\}$. The semantic information of a fixed activity at the destination can be recorded as $sm_DestSemFixedAct_5^{Lisa} = \{StayHome, Home_1^{Lisa}, 20:00, null\}$. The semantic information of the flexible activity to be scheduled can be recorded as $sm_SemFlexibleAct_4^{Lisa} = \{Shopping, 30\ min, expensive, true\}$. Then, the semantic information at the prism level can be maintained as $sm_SemPrism_4^{Lisa} = \{sm_OriSemFixedAct_3^{Lisa}, sm_DestSemFixedAct_5^{Lisa}, sm_SemFlexibleAct_4^{Lisa}, 10\ km^2, 30\ km^2*hour, null\}$. The $SemSTCuboidSet_i^q$ element is determined as a series of $SemSTCuboid_j^{Lisa}$. A typical semantic space-time cuboid near her workplace is $SemSTCuboid_{23}^{Lisa}$ with the set of semantic information as $sm_SemSTCuboid_{23}^{Lisa} = \{5\ min, 25\ min, 150\ min, (workplaceNeighborhood)\}$, where $workplaceNeighborhood$ is the place contexts of the cuboid. Each cuboid consists of a set of semantic space-time cubes. The frequency of visits by Lisa during the past three months is recorded at each space-time cube as semantic information. Through this example, it is obvious that the introduced $SemPrism$ entity is a powerful extension of the classical space-time prism by incorporating semantic information at various granularities, i.e., cube, cuboid, and prism levels.

3.2.4. Semantic space-time lifeline entity

This section describes the semantic space-time lifeline entity (denoted by $SemLifeLine^q$) in the physical space. A $SemLifeLine^q$ object consists of a series of $SemPath$ and $SemPrism$ entities during a typical day. It can be expressed as

$$SemLifeLine^q = \{PhyEntitySet^q, sm_SemLifeLine^q\} \quad (14)$$

where $PhyEntitySet^q$ consists of consecutive $PhyEntity_i^q$ objects, which are the super-type of $SemPath_k^q$ and $SemPrism_i^q$. The set of semantic information at the lifeline level is defined as $\{StartTime, EndTime, Duration, ActivityNum, TotalOutHomeTime, OtherContextSet\}$. We can also project all semantic space-time prisms of the $SemLifeLine^q$ object onto spatial dimensions by removing the time dimension. This projection forms a semantic daily potential path area, denoted by $SemDPPA^q$, which is the union of corresponding $SemPPA_i^q$ entities.

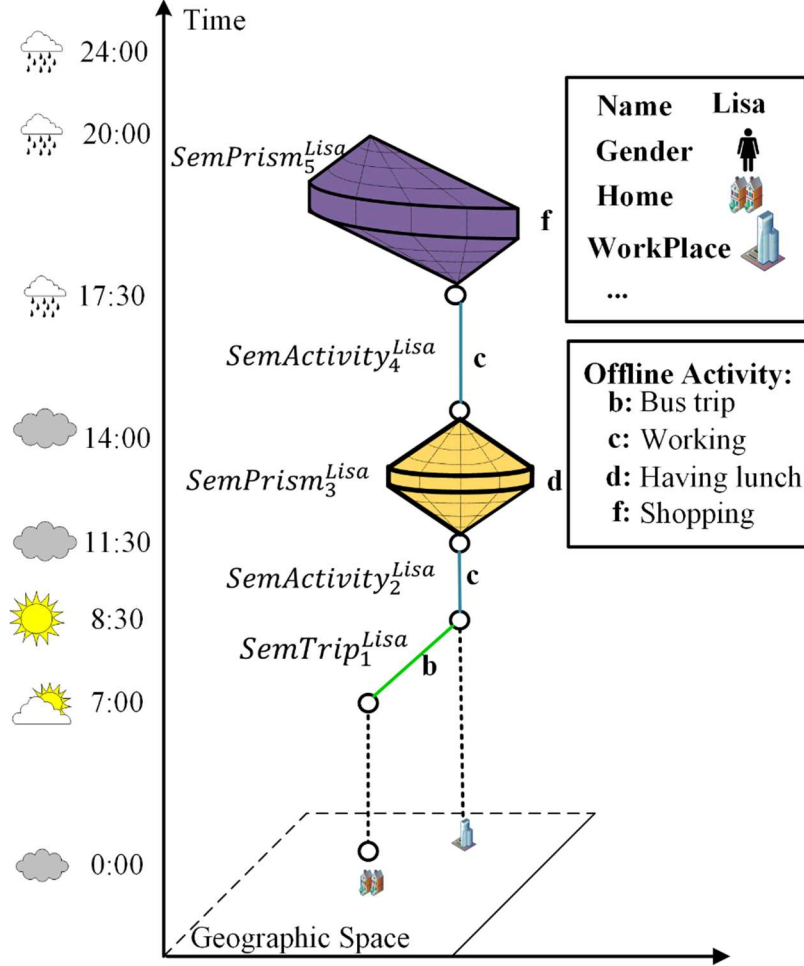


Figure 6. An illustrative example of semantic space-time lifeline.

Figure 6 illustrates a semantic space-time lifeline, $SemLifeLine^{Lisa}$. As shown, this lifeline object consists of a morning commute trip ($SemTrip_1^{Lisa}$), a working activity ($SemActivity_2^{Lisa}$), a prism ($SemPrism_3^{Lisa}$) planning to have lunch, another working activity ($SemActivity_4^{Lisa}$), and another prism ($SemPrism_5^{Lisa}$) planning to do a shopping activity. Accordingly, $PhyEntitySet^{Lisa} = (SemTrip_1^{Lisa}, SemActivity_2^{Lisa}, SemPrism_3^{Lisa}, SemActivity_4^{Lisa}, SemPrism_5^{Lisa})$. The set of semantic information can be recorded as $sm_SemLifeLine^{Lisa} = \{4, 13 \text{ hours}, null\}$.

3.3. Activity-travel behaviors in the hybrid cyber-physical space

This section describes semantic time-geographical entities representing individual activity-travel behaviors in the hybrid cyber-physical space. Figure 7 gives the detailed entities for three semantic time-geographical entities, including semantic website-time paths (in yellow color), semantic hybrid activity patterns (in light blue color), and hybrid projects (in purple color). The descriptions of these entities are given in the following sections.

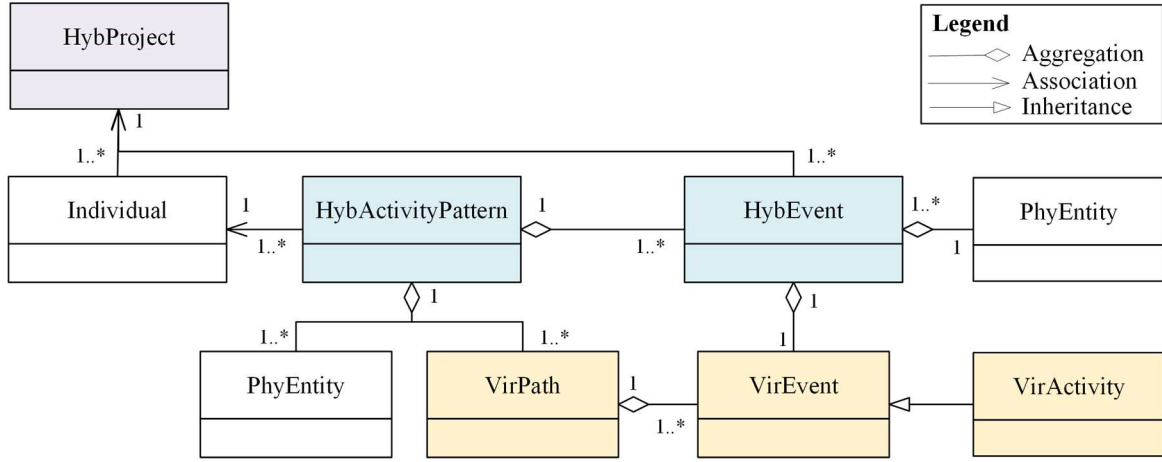


Figure 7. Detailed semantic time-geographical entities in the hybrid cyber-physical space. Note that entities in white color are defined in Figure 3.

3.3.1. Semantic website-time path entity

This section describes the semantic website-time path entity (denoted by $VirPath_i^q$) to represent the individual's online activities in cyberspace (Miller, 2005b). We first introduce a semantic online activity entity (denoted by $VirActivity_i^q$) representing the i^{th} online activity performed by individual q . It can be expressed as

$$VirActivity_i^q = \{sm_VirActivity_i^q\} \quad (15)$$

where $sm_VirActivity_i^q$ is the set of semantic information as $\{ActivityGoal, Website, StartTime, EndTime, Duration, OtherContextSet\}$. The *Website* attribute represents the visited website or application (e.g., Amazon, YouTube, Facebook, and Google) during the online activity. Compared to activities performed in the physical space ($SemActivity_i^q$), this $VirActivity_i^q$ object relaxes space-time constraints of physical presence to the specific facility. We then define $VirPath_i^q$ entity as

$$VirPath_i^q = \{VirActivitySet_i^q, sm_VirPath_i^q\} \quad (16)$$

where $VirActivitySet_i^q$ is the set of online activities performed by the individual. $sm_VirPath_i^q$ is the set of semantic information defined as $\{ActivityNum, Duration, OtherContextSet\}$, where *ActivityNum* represents the number of conducted online activities and *Duration* represents the total time used for online activities. An illustrative example of $VirPath_i^q$ is given in Supplementary Section S6.

3.3.2. Hybrid activity pattern entity

This section introduces the entity of the semantic hybrid activity pattern ($HybActivityPattern_d^q$) for representing individual daily activities in the hybrid cyber-physical space during a typical day d . We first introduce a hybrid event (denoted by $HybEvent_i^q$) entity. It is used to represent an event element with homogeneous activity semantics in the hybrid cyber-physical space, when we combine individual online activities ($VirPath_i^q$) with offline activities ($PhyEntity_i^q$). A $HybEvent_i^q$ object can be expressed as

$$HybEvent_i^q = \{sm_HybEvent_i^q\} \quad (17)$$

where $sm_HybEvent_i^q$ is the set for recording semantic information at the hybrid event level. We further define $sm_HybEvent_i^q = \{OfflineActivity, VirtualActivity, StartTime, EndTime, isHybrid\}$, where *OfflineActivity* maintains a $PhyEntity_j^q$ object that can be an object of $SemActivity_j^q$, $SemTrip_j^q$, or $SemPrism_j^q$; *VirtualActivity* records a $VirActivity_j^q$ object; *StartTime* and *EndTime*

are the starting and ending times of this $HybEvent_i^q$ object; and $isHybrid$ is a Boolean variable indicating whether $VirtualActivity$ is null.

Figure 8 illustrates a simple example by associating a $VirPath_i^q$ object with a $SemPath_k^q$ object. As shown, Lisa performed 7 online activities in the $VirPath_i^q$ object and 11 offline activities in the $SemPath_k^q$ object. The combination of her online and offline activities can increase activity fragmentation by generating 23 $HybEvent_i^{Lisa}$ objects. Among them, 7 $HybEvent_i^{Lisa}$ objects have $isHybrid = true$ in the hybrid cyber-physical space, including online shopping during the commuting trip ($HybEvent_3^{Lisa}$), online entertainment during the commuting trip ($HybEvent_5^{Lisa}$), online information search during the working activity ($HybEvent_8^{Lisa}$), online socializing during the launch ($HybEvent_{12}^{Lisa}$), online socializing at home ($HybEvent_{20}^{Lisa}$), online entertainment and socializing at home ($HybEvent_{21}^{Lisa}$), and online socializing at home ($HybEvent_{22}^{Lisa}$). Other 16 $HybEvent_i^{Lisa}$ objects have $isHybrid = false$ in the physical space.

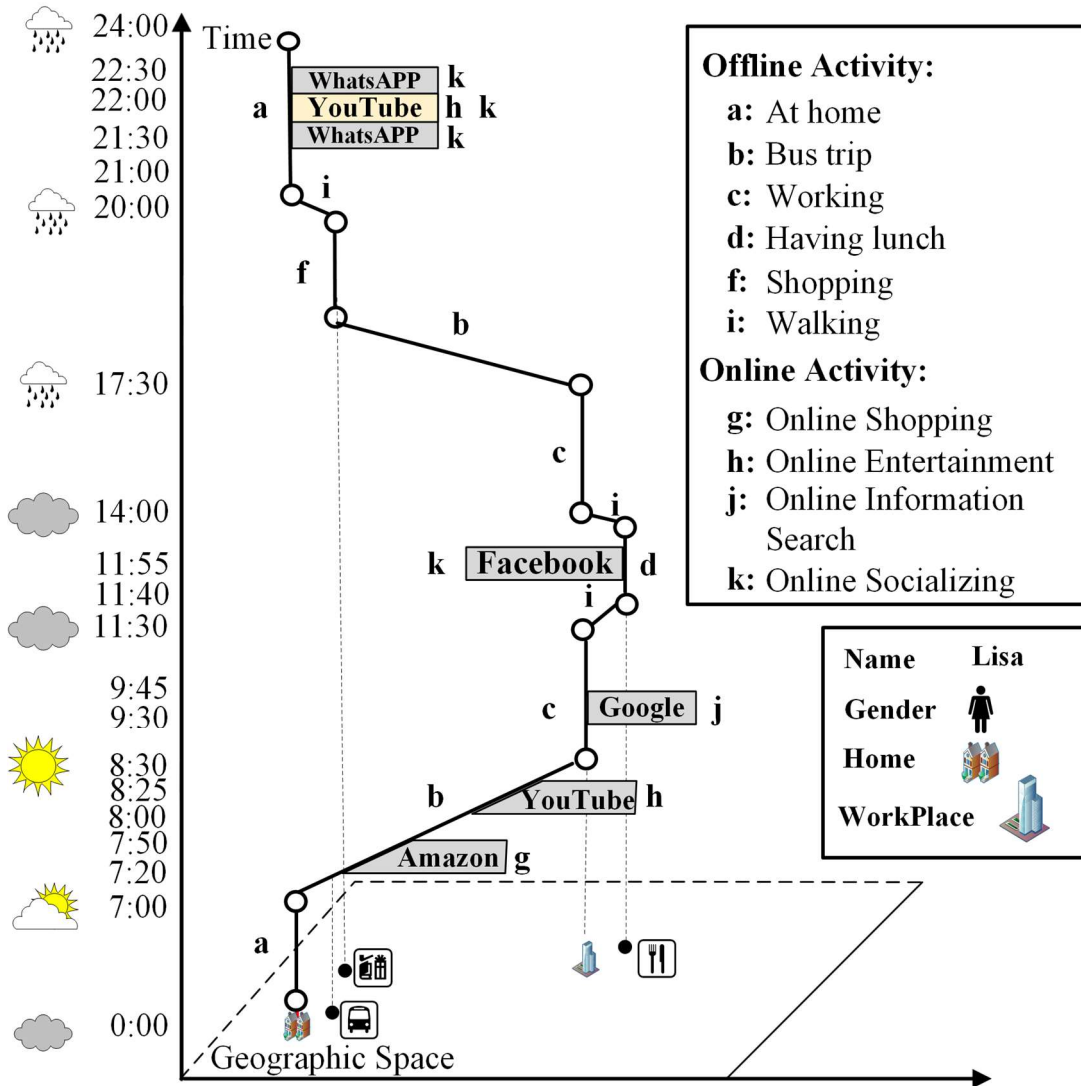


Figure 8. An illustrative example of a semantic hybrid activity pattern entity.

Subsequently, we define the semantic hybrid activity pattern ($HybActivityPattern_d^q$) as

$$HybActivityPattern_d^q = \{VirPath_i^q, PhyEntity_i^q, HbyEventSet_d^q, sm_HybActPattern_d^q\} \quad (18)$$

where $VirPath_i^q$ represents online activities conducted by individual q ; $PhyEntity_i^q$ represents offline activities conducted by individual q ; $HybEventSet_d^q$ is the set of $HybEvent_i^q$ objects by combining $VirPath_i^q$ and $PhyEntity_i^q$ objects; and $sm_HybActivityPattern_d^q$ is the set of semantic information. We further define $sm_HybActivityPattern_d^q = \{Date, WorkdayType, HybElementNum, OtherContextSet\}$, where $Date$ is the date information, $WorkdayType$ is the type of working day (e.g., weekday, weekend, and public holiday), and $HybElementNum$ is the number of $HybEvent_i^q$ objects representing the level of activity fragmentation.

3.3.3. Hybrid Project Entity

This section describes the hybrid project entity (denoted by $HybProject_i^q$). Following time-geographical studies (Ellegård, 2018), a $HybProject_i^q$ object is defined as a series of sequenced activities performed by one or more persons for a common goal. Therefore, it can be expressed as

$$HbyProject_i^q = \{HbyEventSet_d^q, SemIndividualSet_i^q, sm_SemProject_i^q\} \quad (19)$$

where $HbyEventSet_d^q$ is the set of recording online and/or offline activities (i.e., $HybEvent_i^q$ objects) involved in this project. Multiple activities are sorted in chronological order. For each $HybEvent_i^q$, we maintain one corresponding $SemIndividual^q$ (who conducted this activity element) in the $SemIndividualSet_i^q$ element. If only one individual is involved in this project, we can simply set $SemIndividualSet_i^q = null$. The set of semantic information, $sm_SemProject_i^q$, is defined as $\{Description, ActivityElementNum, IndividualNum, OtherContextSet\}$, where $Description$ records the goal of the project, $ActivityElementNum$ records the number of activity elements involved in the project, and $IndividualNum$ represents the number of individuals involved in the project without double counting.

Figure 9 illustrates a project $HybProject_1^{Lisa}$ object established by Lisa for preparing and having dinner at her home with her mother, named Lily. This project involves five activity elements conducted by three individuals. The first one, $HybEvent_1^{Lisa}$, is the online food purchase activity performed by Lisa during her commuting trip. The second one, $HybEvent_2^{Mike}$, is the food delivery activity to Lisa's home was conducted by Mike, an employee of the online shopping company. The third one, $HybEvent_3^{Lily}$, is conducted by Lily to cook food at home. The fourth and fifth ones, $HybEvent_4^{Lisa}$ and $HybEvent_5^{Lily}$, are joint activities conducted by Lisa and her mother to have dinner together at home. Therefore, the $HbyEventSet_d^q$ set maintains all five $HybEvent_i^q$ objects; and the $SemIndividualSet_i^q$ set maintains corresponding five individuals as $\{Lisa, Mike, Lily, Lisa, Lily\}$. The semantic information of this project can be recorded as $\{\text{"preparing and having dinner at home"}, 5, 3, null\}$.

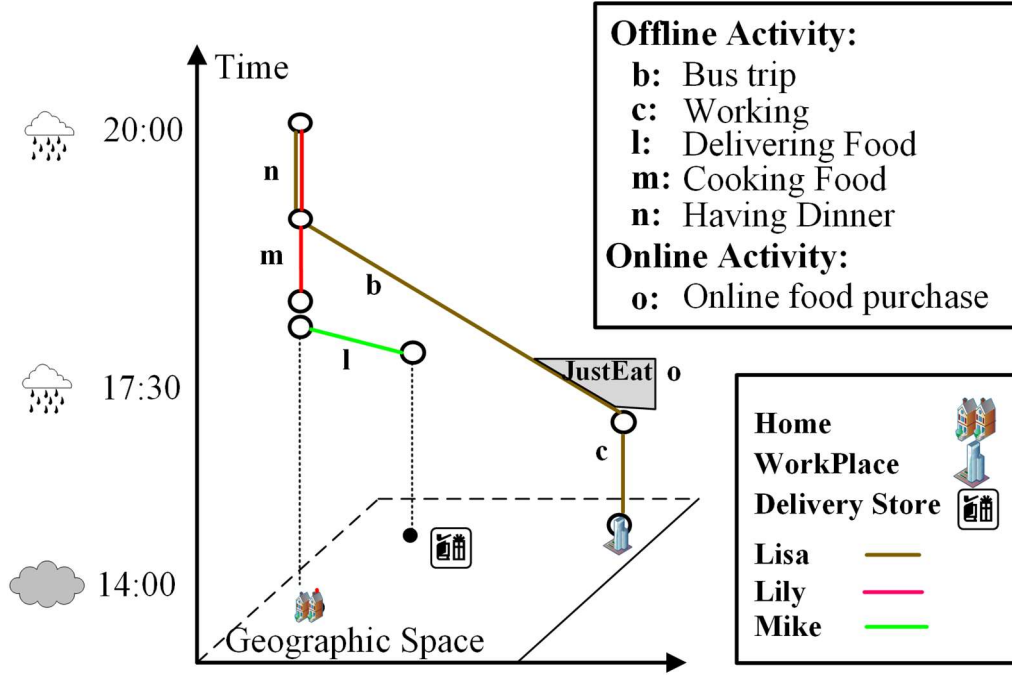


Figure 9. An illustrative example of a project in the hybrid physical-virtual space.

4. Semantic time-geographical relationships defined in the social space

This section describes two types of semantic time-geographical relationships defined in the social space: semantic social relationships and semantic space-time relationships. These two types of relationships are described in the following sections.

4.1. Semantic social relationships

This section describes semantic social relationships. Given two individuals, $SemIndividual^p$ and $SemIndividual^q$, we can define their household membership as $\delta_{household}(SemIndividual^p, SemIndividual^q) = 1$ if they are within the same household $SemHousehold^h$, and $\delta_{household} = 0$ otherwise. Similarly, we can define the colleague relationship $\delta_{colleague}(SemIndividual^p, SemIndividual^q) = 1$ if they are working in the same company $SemCompany^c$, and $\delta_{colleague} = 0$ otherwise. We can also define other types of social relationships between two individuals according to the applications on hand, such as friendships. The incorporation of social relationships into time-geographical studies can better schedule or understand activity-travel behaviors among individuals.

4.2. Semantic space-time relationships

This section describes three semantic space-time relationships, including semantic space-time bundling, semantic space-time intersection, and mediated bundling relationships. We first introduce the semantic space-time bundling relationship, denoted by δ_{Bundle} . In the classical time-geographical framework, bundling refers to the convergence of two or more paths by considering only space and time dimensions. It is necessary to further consider semantic dimensions in the identification of the bundling relationship. As illustrated in Figure 10, there are five semantic space-time paths ($SemActivity_5^{Lisa}$, $SemActivity_5^{Marry}$, $SemActivity_4^{John}$, $SemActivity_4^{Lily}$, $SemActivity_4^{Emma}$) clustered during 19:00-20:00 at a shopping mall. Among these paths, the first three paths have $ActivityGoal = shopping$, but the last two paths have $ActivityGoal = working$. If only space and time dimensions are considered, all five paths are identified within a single bundle. It is more reasonable to further consider the activity type to identify two distinctive bundles for

shopping and working, respectively. Following this idea, we refine the bundling relationship between two semantic space-time paths $SemPath_i^q$ and $SemPath_j^p$ as:

$$\delta_{Bundle}(SemPath_i^q, SemPath_j^p) = \delta_{Bundle}^{Geo}(SemPath_i^q, SemPath_j^p) \delta_{Bundle}^{Sem}(SemPath_i^q, SemPath_j^p) \quad (20a)$$

$$\delta_{Bundle}^{Geo}(SemPath_i^q, SemPath_j^p) = \begin{cases} 1, & \text{if } GeoD(SemPath_i^q(t), SemPath_j^p(t)) \leq \omega_G, \forall t \in [t_s, t_e] \\ 0, & \text{otherwise} \end{cases} \quad (20b)$$

where δ_{Bundle} is a binary function, $\delta_{Bundle} = 1$ indicates a semantic space-time bundling relationship and $\delta_{Bundle} = 0$ otherwise. It can be decomposed into two binary components. The $\delta_{Bundle}^{Geo}(SemPath_i^q, SemPath_j^p)$ component is the corresponding space-time relationship defined from the space-time distance perspective, indicating that the spatial distance $GeoD()$ between two paths at every time instant t should be less than a given threshold ω_G during the period $[t_s, t_e]$. The $\delta_{Bundle}^{Sem}(SemPath_i^q, SemPath_j^p)$ component is the corresponding activity semantic relationship defined from a semantic perspective. In this example, $\delta_{Bundle}^{Sem} = 1$ if two individuals have the same activity goal.

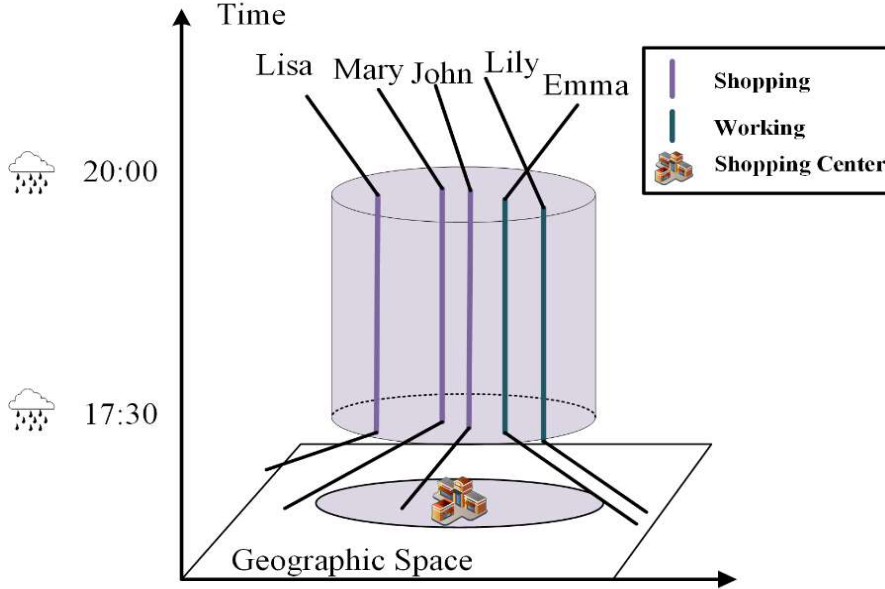


Figure 10. Example of bundling relationships among five semantic space-time paths.

We then introduce the semantic space-time intersection relationship, denoted by $\delta_{Intersect}$. In the classical time-geographical framework, a space-time intersection relationship between two time-geographical entities represents the scenario of two entities coexisting in space and time dimensions. This relationship could be further enhanced by incorporating semantic information. For example, the space-time intersection of two space-time prisms is commonly used to determine the feasibility of scheduling a joint activity between two individuals (Farber *et al.* 2013). As illustrated in Figure 11, by incorporating semantic information, the joint activity can be better scheduled by considering not only the feasible activity locations but also the similarity of activity goals. Accordingly, the semantic space-time intersection of two prisms can be defined as

$$\delta_{Intersect}(SemPrism_i^q, SemPrism_j^p) = \delta_{Intersect}^{Geo}(SemPrism_i^q, SemPrism_j^p) \delta_{Intersect}^{Sem}(SemPrism_i^q, SemPrism_j^p) \quad (21a)$$

$$\delta_{Intersect}^{Geo}(SemPrism_i^q, SemPrism_j^p) = \begin{cases} 1, & \text{if } SemPrism_i^q(t) \cap SemPrism_j^p(t) \neq \emptyset, \exists t \\ 0, & \text{otherwise} \end{cases}$$

(21b)

where $\delta_{Intersect}$ is a binary function, $\delta_{Intersect} = 1$ indicates a semantic space-time intersection relationship between two prisms and $\delta_{Intersect} = 0$ otherwise. The $\delta_{Intersect}$ function can be divided into two components. The $\delta_{Intersect}^{Geo}(SemPrism_i^q, SemPrism_j^p)$ component represents the corresponding space-time relationship defined from space and time dimensions. The $\delta_{Intersect}^{Sem}(SemPrism_i^q, SemPrism_j^p)$ is the corresponding activity semantic relationship defined from a semantic perspective.

Similarly, the semantic space-time intersection relationship can also be defined for other types of geographical entities (i.e., prism-path, prism-station, and path-path intersections). For example, the $\delta_{Intersect}(SemPrism_i^q, SemStation_j^f)$ relationship is applied to introduce a new semantic accessibility measure in the following sections.

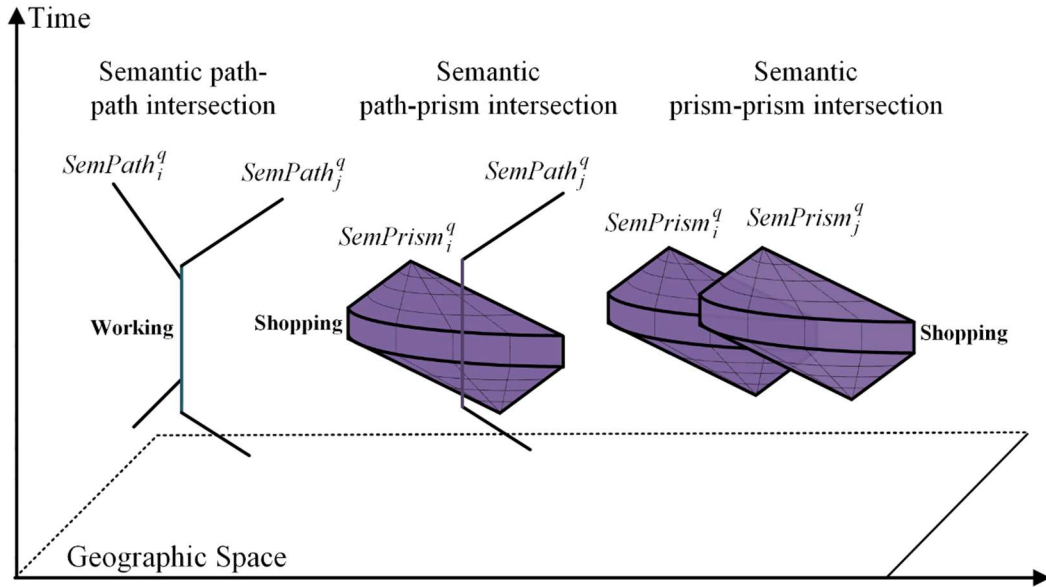


Figure 11. An illustrative example of semantic space-time intersection relationships among different semantic space-time entities in the physical space.

We further introduce the mediated bundling relationship between two online activities (Thulin et al., 2020; Shaw, 2023; Li et al., 2024). Let $\delta_{Bundle}^{Sem}(VirActivity_i^q, VirActivity_j^p)$ be the mediated bundling relationship between two online activities, $VirActivity_i^q$ and $VirActivity_j^p$. We can determine $\delta_{Bundle}^{Sem} = 1$ if two individuals are mediated togetherness, and $\delta_{Bundle}^{Sem} = 0$ otherwise. For example, direct social contacts via WhatsApp indicate the mediated togetherness of $VirActivity_i^q$ and $VirActivity_j^p$.

5. Implementation and accessibility application

5.1. Relational database implementation

This section describes relational database approaches to implement the proposed framework of semantic time geography. We employed the space-time cube model (Nakaya, 2013) for framework implementation. As illustrated in Figure 12, the space-time cube model is a discrete organization of space-time coordinate space with geographical horizontal axes and a temporal vertical axis. Similar to the classical 2D raster data, the whole space-time coordinate space is divided into a set of 3D space-time cubes. The size of a space-time cube (e.g., 500m $\times$ 500m $\times$ 5 min) represents the corresponding space-time resolution. Each space-time cube, denoted by $GeoSTCube_i^q$, can be

expressed as $\langle rowID_i^q, columnID_i^q, timeID_i^q \rangle$, where $rowID_i^q$, $columnID_i^q$ and $timeID_i^q$ respectively are the IDs (i.e., coordinates) in the space-time coordinate space. Using this data model, all semantic time-geographical entities are discretized into a set of 3D space-time cubes. The spatiotemporal data model of semantic time-geographical entities can be transferred into the logic model in the relational database, shown in Supplementary Section S7 using the entity-relationship (ER) diagram.

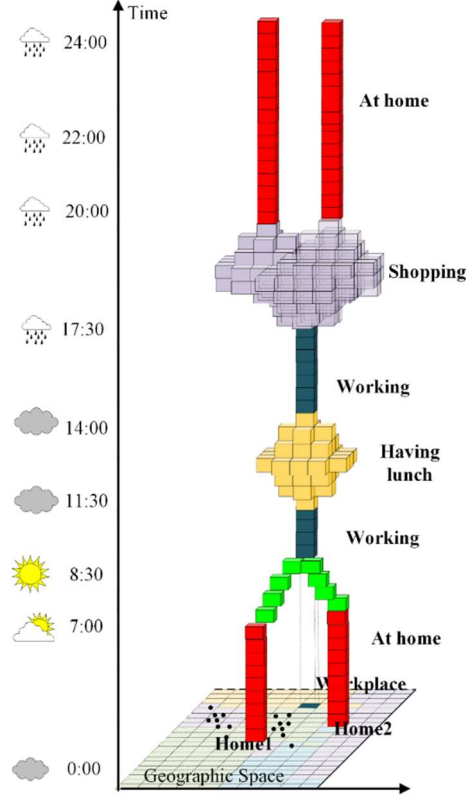


Figure 12. Example of representing two semantic lifeline entities in the space-time cube model.

It is easy to integrate the relational database with existing GIS platforms (e.g., ArcGIS Pro), which are powerful for visualizing and analyzing the semantic time-geographical entities and relationships in space and time dimensions. This study utilized ArcGIS Pro, which employs the NetCDF (Network Common Data Form) format (ESRI, 2023) to organize data as a time series of raster layers. Therefore, every $GeoSTCube_i^q$ can be matched onto data in the NetCDF format by setting $rowID_i^q$ and $columnID_i^q$ on the corresponding cell of the raster and $timeID_i^q$ on the corresponding layer. Figure 13 illustrates the matched NetCDF data for visualizing a semantic lifeline shown in Figure 12. In this example, we set different values at corresponding space-time cubes for visualizing activity types in different colors. Values of 1, 2, and 3, respectively, represent at-home, having lunch, and shopping activities. The user interface of the implemented framework is given in Supplementary Section S8.

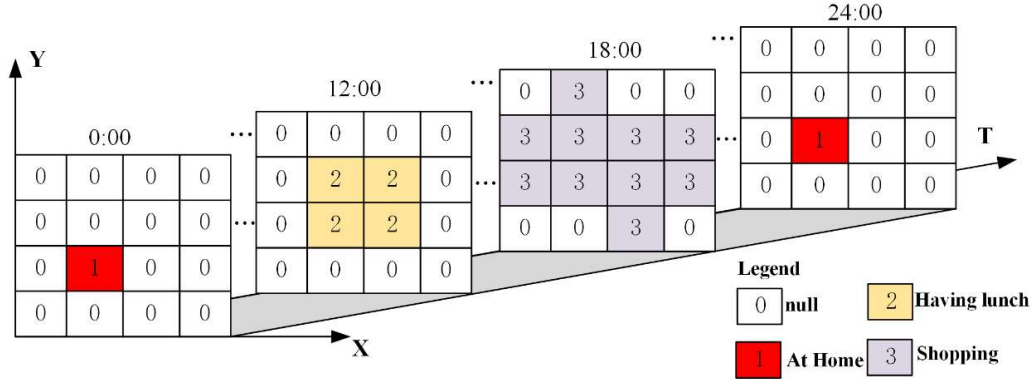


Figure 13. Example of using ArcGIS Pro to visualize a semantic lifeline.

5.2. Semantic accessibility analysis considering semantic contexts

To illustrate the applicability of the proposed framework, we introduce a semantic accessibility measure by incorporating semantic contexts into the conventional measure of individual accessibility (Kwan and Weber, 2003). The introduced measure evaluates how easily individuals can reach preferred facilities, given their activity goals and/or preferences. Given a semantic space-time prism entity of individual q (i.e., $SemPrism_i^q$), the introduced measure (denoted by $SemCUM_i^q$) quantifies the individual accessibility by counting the number of reachable facilities, i.e., semantic space-time station entities satisfying the semantic space-time intersection relationship $\delta_{Intersect}(SemPrism_i^q, SemStation_j^f) = 1$. Each accessible facility $SemStation_j^f$ should not only be reachable in space-time dimensions satisfying $\delta_{Intersect}^{Geo}(SemPrism_i^q, SemStation_j^f) = 1$ but also meet the individual's activity goals and/or preferences satisfying $\delta_{Intersect}^{Sem}(SemPrism_i^q, SemStation_j^f) = 1$ in semantic dimensions. The introduced measure is formally expressed as

$$\begin{aligned}
 SemCUM_i^q &= \sum_{\forall SemStation_j^f} \delta_{Intersect}(SemPrism_i^q, SemStation_j^f) \\
 &= \sum_{\forall SemStation_j^f} \delta_{Intersect}^{Geo}(SemPrism_i^q, SemStation_j^f) \delta_{Intersect}^{Sem}(SemPrism_i^q, SemStation_j^f) \quad (22a)
 \end{aligned}$$

$$\delta_{Intersect}^{Geo}(SemPrism_i^q, SemStation_j^f) = \begin{cases} 1, & \text{if } SemPrism_i^q(t) \cap SemStation_j^f(t) \neq \emptyset, \exists t \\ 0, & \text{otherwise} \end{cases} \quad (22b)$$

This introduced measure is a generalization of the conventional individual accessibility measure (denoted by $GeoCUM_i^q$) (Kwan and Weber, 2003), which considers only $\delta_{Intersect}^{Geo}$ component instead of $\delta_{Intersect}^{Sem}$. The $\delta_{Intersect}^{Sem}$ component can be defined in different ways to incorporate various semantic contexts from both individual and facility sides. One of the simplest way could be matches between activity goals and service types of facilities as below:

$$\delta_{Intersect}^{Sem}(SemPrism_i^q, SemStation_j^f) = \begin{cases} 1, & ActivityGoal_i^q = ServiceType_j^f \\ 0, & ActivityGoal_i^q \neq ServiceType_j^f \end{cases} \quad (23)$$

6. Case Study

6.1. Data collections

A well-developed relational database, i.e., MySQL, was employed to implement the semantic time-geographical framework. Two datasets were collected for the case study. The first dataset was a small open dataset of activity-diary surveys of COVID-19 cases in Wuhan, China. During the early stage of the pandemic, activity-diary surveys were conducted in Wuhan City to record the trajectories and

activity participation of COVID-19 cases. In this study, activity-diary data of 44 individuals were collected from a local website of Wuhan (<http://wh.bendibao.com>) between 20 July and 10 August 2021. The detailed description of this dataset is given in Supplementary Section S9.

The second dataset consists of simulated activity-travel behavior data of 10,970 individuals within 5,000 households. We utilized the data collected from the 44 individuals in the first dataset as templates. A procedure is developed to simulate the data, and its detailed description is given in Supplementary Section S10.

6.2. Visual analysis results

The first dataset with activity-diary data of 44 individuals was utilized for visual analysis. Figure 14(a) visualizes three entities of semantic hybrid activity patterns (i.e., $HybActivityPattern_d^{John}$, $HybActivityPattern_d^{Nancy}$ and $HybActivityPattern_d^{Daisy}$) for the three household members in Family #1 during a typical workday (i.e., 2 August 2021). As shown, the family chose the residential location near the wife's (i.e., Nancy's) workplace. The husband (i.e., John) commuted a long distance in the morning by using the bus. He spent a long out-of-home activity time (i.e., >11 hours) and conducted five out-of-home activities. Semantic space-time prisms (i.e., $SemPrism_{BRK}^{John}$, $SemPrism_{LUN}^{John}$ and $SemPrism_{DIN}^{John}$) based on the walking mode were constructed for three activities (i.e., having breakfast, lunch, and dinner) because they were flexible activities and no activity location was provided in the dataset for these three activities. Compared to her husband, the wife spent less out-of-home activity time (i.e., <7 hours) and took on more household duties, including three joint activities with her daughter (i.e., Daisy) and one fresh food purchase activity. A semantic space-time prism based on the private car mode (i.e., $SemPrism_{PFF}^{Nancy}$) was constructed for the fresh food purchase activity. It can be observed that the size of this semantic space-time prism was much larger than that of the three prisms of her husband using the walking mode. This is because this family only had one car, and it was used by the wife. It reveals a typical pattern of duty and resource allocations in many households in China and beyond (Chen et al., 2011). Therefore, this illustrative example highlights the capabilities of the proposed semantic time-geographical framework for representing both geometric and semantic perspectives of human activity-travel behaviors in physical space.

For demonstration, we added four online activities for both household heads. John conducted online socializing ($VirActivity_1^{John}$) and online shopping ($VirActivity_2^{John}$) during his commuting trip. As shown in Figure 14(d), he first made social contact via WeChat with his wife and then conducted online shopping via Taobao. Nancy made an online socializing ($VirActivity_1^{Nancy}$) and teleworking ($VirActivity_2^{Nancy}$) at home. It formed a mediated bundling relationship, $\delta_{Bundle}^{Sem}(VirActivity_1^{John}, VirActivity_1^{Nancy}) = 1$. Therefore, this illustrative example demonstrates the capabilities of the proposed framework for representing individual activities in the hybrid cyber-physical space.

Figure 14(c) visualizes a semantic space-time intersection relationship between two paths of the wife and daughter from 13:25 to 14:30, i.e., $\delta_{Intersect}(SemPath_i^{Nancy}, SemPath_j^{Daisy}) = 1$. As can be seen, these two paths not only intersected at the space and time dimensions (i.e., $\delta_{Intersect}^{Geo}(SemPath_i^{Nancy}, SemPath_j^{Daisy}) = 1$) but also had the same activity goal (i.e., $\delta_{Intersect}^{Sem}(SemPath_i^{Nancy}, SemPath_j^{Daisy}) = 1$). This relationship was the result of a joint activity between the wife and daughter. Figure 14(b) visualizes a classical space-time intersection relationship between two prisms of the couple from 17:00 to 19:00. As shown, two prisms intersected at the space

and time dimensions, i.e., $\delta_{Intersect}^{Geo}(SemPrism_i^{Nancy}, SemPrism_j^{John}) = 1$. However, they have different activity goals with $\delta_{Intersect}^{Sem}(SemPrism_i^{Nancy}, SemPrism_j^{John}) = 0$. Accordingly, the semantic space-time intersection relationship did not hold, i.e., $\delta_{Intersect}(SemPrism_i^{Nancy}, SemPrism_j^{John}) = 0$; and it cannot be feasible to schedule a joint activity between the couple during this period. Therefore, compared to the classical time-geographical relationships, the introduced semantic time-geographical relationships can be used to analyze human activity-travel behaviors more accurately by incorporating rich contextual information about activities.

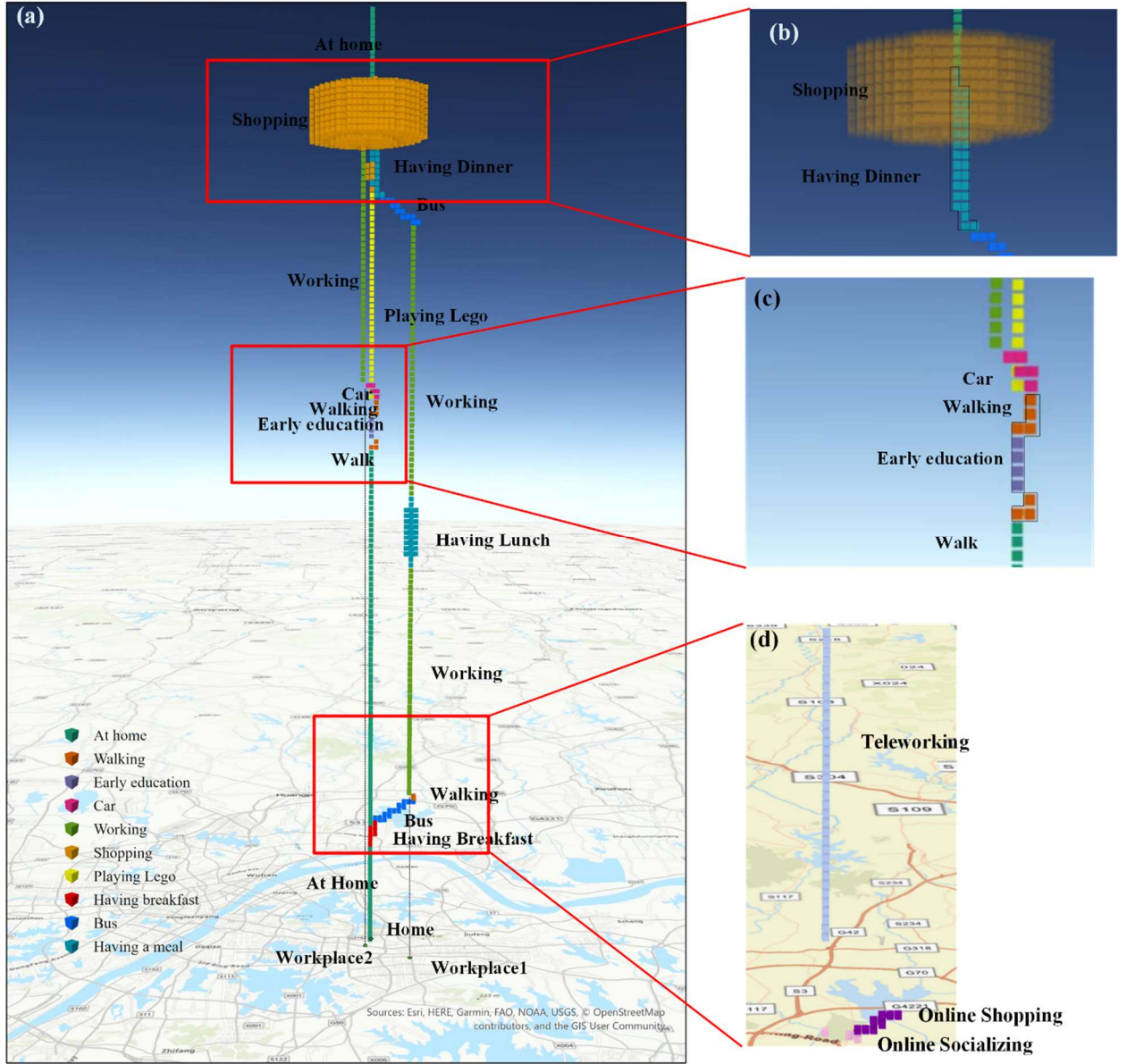


Figure 14. Three entities of semantic hybrid activity patterns for the three household members in Family #1: (a) semantic time-geographical entities; (b) classical space-time intersection relationship between two prisms; (c) semantic space-time intersection relationship between two paths; (d) activities in the hybrid cyber-physical space.

6.3. Semantic accessibility analysis results

Using the introduced semantic accessibility measure, we evaluated accessibility to food retailers for two household heads of Family #1 (i.e., John and Nancy) based on their semantic space-time lifeline

entities (i.e., $SemLifeLine^{John}$ and $SemLifeLine^{Nancy}$) shown in Figure 14. As shown in the figure, $SemLifeLine^{John}$ consisted of three semantic space-time prisms (i.e., $SemPrism_{BRK}^{John}$, $SemPrism_{LUN}^{John}$ and $SemPrism_{DIN}^{John}$) for scheduling his breakfast, lunch, and dinner, respectively. $SemLifeLine^{Nancy}$ had only one semantic space-time prism (i.e., $SemPrism_{PFF}^{Nancy}$) for scheduling fresh food purchase activity in supermarkets or vegetable markets after work. Accordingly, the food accessibility for household heads was calculated by checking semantic space-time intersection relationships, i.e., $\delta_{Intersect}(SemPrism_i^q, SemStation_j^f)$, between corresponding prisms and food retailers in ten different categories.

It should be noted that food retailers of different categories had distinct service types to meet individual activity goals, i.e., varying $\delta_{Intersect}^{Sem}(SemPrism_i^q, SemStation_j^f)$ values among retailer categories. According to previous work (Wang et al., 2018), we set $\delta_{Intersect}^{Sem} = 1$ for the breakfast goal to food retailers in three categories: Bakery and beverage stores, Fast-food outlets, and Noodle houses. For the lunch goal, we set $\delta_{Intersect}^{Sem} = 1$ to retailers in four categories: Fast-food outlets, Noodle houses, Chinese restaurants, and Foreign restaurants. For the dinner goal, we set $\delta_{Intersect}^{Sem} = 1$ to retailers in four categories: Chinese restaurants, Foreign restaurants, Hot-pot restaurants, and Barbecue shops. For the goal of fresh food purchases, we set $\delta_{Intersect}^{Sem} = 1$ to retailers in two categories: Supermarkets, and Fruit and vegetable shops. We set $\delta_{Intersect}^{Sem} = 0$ to retailers in the category of Convenience stores for all four activity goals, because they cannot provide appropriate service to fulfill these activity goals.

Figure 15(a) gives the results of the semantic accessibility analysis for John. As shown in Figure 15(b), $SemCUM_{BRK}^{John} = 17$ was obtained, indicating that 17 retailers near his home location (i.e., 4 Bakery and beverage stores, 9 Fast-food outlets, and 4 Noodle houses) are feasible for John to schedule breakfast. As can be seen in Figure 15(c), $SemCUM_{LUN}^{John} = 4$ was determined, showing that 4 retailers surrounding his workplace (i.e., 3 Fast-food outlets and 1 Chinese restaurant) are accessible for John to have lunch. As shown in Figure 15 (d), $SemPrism_{DIN}^{John} = 19$ was obtained, indicating that 19 retailers (i.e., 15 Chinese restaurants, 1 Foreign restaurant, 1 Hot-pot restaurant, and 2 Barbecue shops) are accessible for John to have dinner together with his colleagues. This result underscores that individual accessibility to food retailers can be varying at different times of the day to meet distinct goals of food consumption (Wang et al., 2018). By summing all three above accessibility results, the total semantic accessibility of John is $SemCUM_{total}^{John} = 40$.

Figures 15(e) and 15(f) give the results of the semantic accessibility analysis for Nancy. Compared to her husband, Nancy had a much higher level of accessibility to retailers, i.e., $SemCUM_{PFF}^{Nancy} = 288$. This result is consistent with the previous observation that household heads with cars tend to have higher levels of food accessibility than the other heads without cars (Smith et al., 2022). Further, we found that the female head could have access to different categories of food retailers (i.e., Supermarkets, and Fruit and vegetable shops) compared to the male head, because of their different roles in the household. Accordingly, the total food accessibility for Family #1, $SemCUM_{Total}^{Household} = 328$, can be easily calculated by adding $SemCUM_{PFF}^{Nancy}$ and $SemCUM_{total}^{John}$.

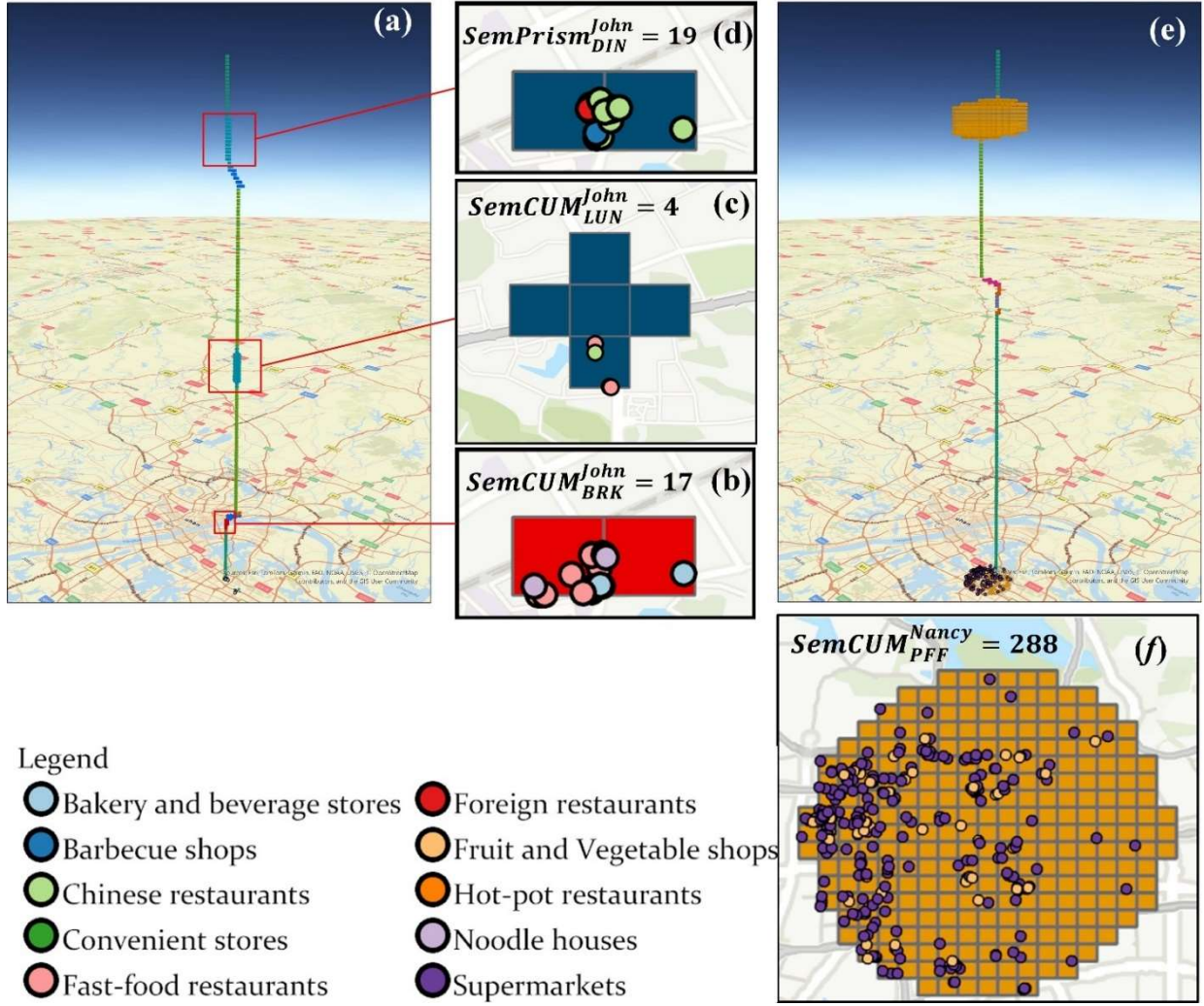


Figure 15. Semantic accessibility analysis results of John and Nancy considering semantic contexts: (a) $SemLifeLine^{John}$; (b) $SemCUM_{BRK}^{John}$; (c) $SemCUM_{LUN}^{John}$; (d) $SemPrism_{DIN}^{John}$; (e) $SemLifeLine^{Nancy}$; (f) $SemCUM_{PFF}^{Nancy}$.

Figure 16 gives the results of the classical accessibility analysis ignoring semantic contexts. In this case, retailers in all ten categories are considered for evaluating the accessibility of both John and Nancy. Compared to the above results of semantic accessibility, we observed a significantly higher level of classical accessibility. As shown in the figure, the classical accessibility of Nancy was $GeoCUM_{PFF}^{Nancy} = 3,107$, i.e., about 9.8 times higher than $SemCUM_{PFF}^{Nancy}$; while the classical accessibility of John was $GeoCUM_{total}^{John} = 105$, about 1.6 times higher than $SemCUM_{total}^{John}$. Therefore, the introduced semantic accessibility measure extends the classical measure of individual accessibility by explicitly considering semantic matches between individual activity goals and service types of urban facilities.

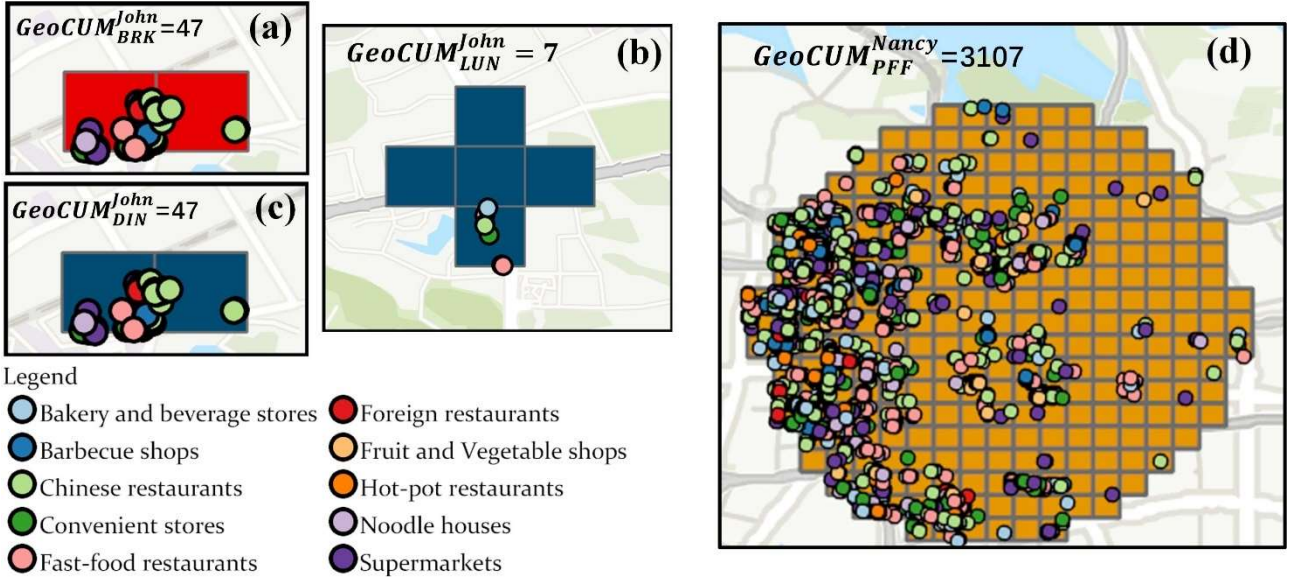


Figure 16. Classical accessibility analysis results of Nancy and John without consideration of semantic contexts: (a) $GeoCUM_{BRK}^{John}$; (b) $GeoCUM_{LUN}^{John}$; (c) $GeoCUM_{DIN}^{John}$; (d) $GeoCUM_{PFF}^{Nancy}$.

6.4. Query results

We utilized the second dataset with simulated activity-travel data of 5,000 households for querying performance analysis. The sizes of 29 tables in this dataset are summarized in Table S3 in Supplementary Section S11. All experiments were conducted on a desktop with a four-core Intel 2.6 GHz CPU (only a single processor was used) and 16 GB of RAM on a Windows 10 operating system.

Using a relational database, we employed Structured Query Language (SQL) to perform various queries on semantic time-geographical entities. In this case study, four types of representative queries were implemented, including entity-based, semantic-based, space-time-based, and relationship-based queries. The following representative queries of these four types were used for performance testing, and their corresponding SQL implementations and user interfaces are given in Supplementary Section S11.

Query (a): Where is John and what is he doing at 12:00?

Query (b): Who are the friends of individuals whose monthly income exceeds 10,000 RMB?

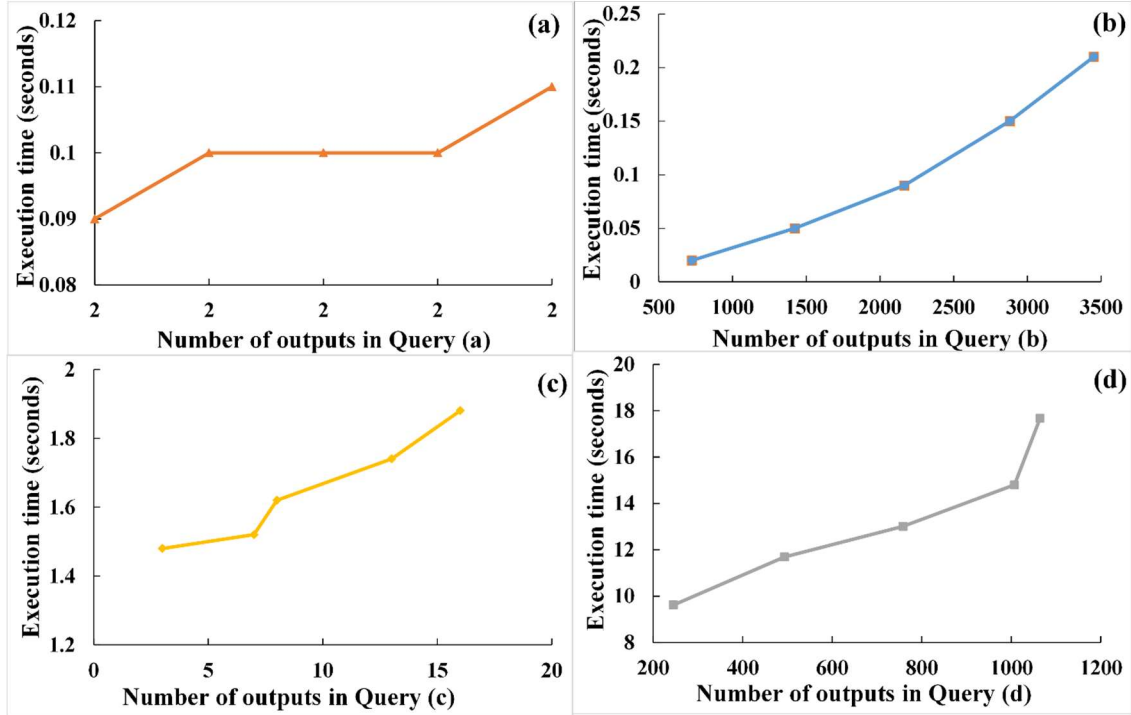
Query (c): Who is likely to have lunch with John?

Query (d): Who visited the region within 2km of the Shopping Center “QunGuang” from 18:00 to 19:00?

The performance of Queries (a)-(d) was investigated under different numbers of households. As shown in Table 1, Query (a) performed well (about 0.1s) under various numbers of households. It is expected, since this query only retrieved two semantic time-geographical entities under various database sizes, see Figure 17(a). The performance of Query (b) degraded with the database size. For example, when the household number was set as 1,000, it required only 0.02s. When household numbers increased to 5,000, the execution time rose to 0.21s. This is because the number of retrieved individuals increased by 4.8 times from 724 to 3450, see Figure 17(b).

Table 1. Execution times in seconds for four queries under different numbers of households.

Query Type	Number of Households				
	1000	2000	3000	4000	5000
Query (a)	0.1	0.1	0.11	0.11	0.12
Query (b)	0.02	0.05	0.09	0.15	0.21
Query (c)	1.48	1.52	1.62	1.74	1.88
Query (d)	9.62	11.69	13.01	14.8	17.67

**Figure 17.** Execution time for the four illustrative queries with different numbers of households and their corresponding output results.

Query (c) consumes more execution time than Queries (a) and (b). This result is expected. Query (c) first retrieved a semantic space-time prism representing the lunch activity of John. The retrieved prism consisted of 227 GeoSTCubes. Query (c) then performed queries on each GeoSTCubes to search for individuals who have a semantic space-time intersection relationship with the retrieved prism. It can be seen that the query performance of Query (c) degraded slightly with the increase in database size. This is because the number of retrieved individuals was not significantly increased with the database size, see Figure 17(c).

Query (d) performed the worst among the four queries. This is because Query (d) determined too many GeoSTCubes (i.e., 1,600) in the constructed space-time region. This query needs to perform queries on all GeoSTCubes to retrieve individuals who intersected the constructed space-time regions. It can be observed that the query performance of Query (d) degraded rapidly with the increase in database size. This result is due to the rapid increase in the number of retrieved individuals with the rise in database size, see Figure 17(d).

We further investigated the performance of Query (d) under different numbers of retrieved *GeoSTCubes* by using different thresholds for constructing buffer zones as 200m, 1km, 1.5km, 2km, and 2.5km. As shown in Figure 18, the query performance was degraded significantly with the increase in the number of *GeoSTCubes*. For example, the execution time was only 1.27s when the number of *GeoSTCubes* was 100. This execution time was increased by 22 times (i.e., $27.93/1.21 - 1$)

when the number of *GeoSTCubes* was increased to 2,500.

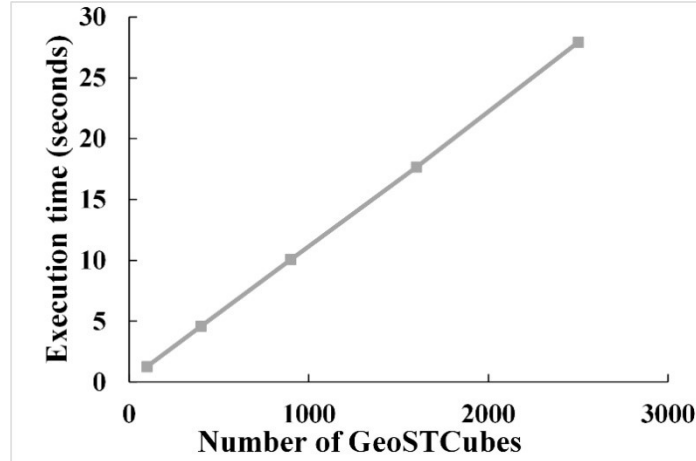


Figure 18. Execution time of Query (d) under different sizes of space-time regions.

7. Conclusion

Although the classical notation system of time geography was powerful in analyzing the geometric aspect of activity-travel behaviors in the space-time dimensions, it neglects the vital semantic contexts, such as activity goals, subjective perceptions and emotions, and environmental conditions. To fill the gap, we proposed a novel framework of semantic time geography by integrating semantic contexts into the notation system of time geography. Along the line of the classical notation system (Miller, 2005a), the proposed framework introduced a set of semantic time-geographical entities (i.e., semantic space-time paths, stations, prisms, and lifelines) to represent semantically enriched activity-travel behaviors in the physical space; and set of semantic space-time intersection relationships by considering not only the coexistence of entities in space-time dimensions but also the match of activity semantics. Further, the proposed framework offered a new set of semantic time-geographical entities (i.e., semantic website-time paths, semantic hybrid activity patterns, and hybrid projects) to represent complicated activity-travel behaviors in the hybrid cyber-physical space. Therefore, the proposed framework provides a generalizable and computable approach for understanding not only space-time but also inherent cognitive dimensions of activity-travel behaviors in hybrid cyber-physical spaces.

The proposed framework was implemented by using the space-time cube model based on the well-developed relational database. This makes it easy to integrate with most existing GIS platforms. To demonstrate the applicability of the proposed framework, a new semantic accessibility measure was introduced by using semantic space-time intersection relationships between prisms and stations. This introduced measure evaluates how easily individuals can access preferred destinations, given their activity goals. It generalizes the conventional individual accessibility measure (Kwan and Weber, 2003) by considering not only the feasibility of activity participation in space-time dimensions but also destination preferences to meet individuals' activity goals.

A case study was carried out using real activity-diary data. Case study results highlighted the capabilities of the proposed framework for representing, visualizing, querying, and analyzing both space-time and semantic dimensions of human activity-travel behaviors in the hybrid cyber-physical space. The introduced semantic accessibility measure can well quantify distinctive patterns of food accessibility for different household members within families, given their different household roles and activity goals. The conventional individual accessibility measure tended to overestimate the food accessibility levels of household members, due to the ignorance of their actual activity goals.

This study was the first to develop the notation system of semantic time geography. Several directions for future research are worth noting. Firstly, this study proposed a comprehensive framework for modeling most of the key concepts in time geography. However, the proposed framework cannot model all concepts in time geography, particularly two complicated concepts, i.e., diorama and pockets of local orders (Ellegård, 2018). By modeling rich semantic contexts of individual activity-travel behaviors, the proposed framework could be a good starting point for modeling these complicated time-geographical concepts. Secondly, we introduced a semantic accessibility measure for illustrating the advantages of incorporating semantic contexts into time-geographical studies. However, the case study only considers the semantic matches between individuals' activity goals and facilities' service types. Further studies are thus required to conduct more comprehensive accessibility studies by considering rich semantic contexts from both individual and facility sides, such as individual perceptions (Dodge and Nelson, 2023), food qualities and prices, and online food purchases (Chen et al., 2025). Thirdly, this study implemented the proposed framework by using the relational database. In recent years, several non-SQL graph databases have been developed, such as Neo4J, which are more flexible to represent the high heterogeneity of contextual information and more efficient for managing big datasets. How to implement the proposed framework using emerging graph databases is another important topic for further investigation. Third, the introduced semantic time-geographical entities were implemented using the space-time cube model. Entities in this implementation can be interpreted as time-geographical fields (Miller and Bridwell, 2009), which apply to both planar and network spaces. However, to present the essential idea, we simply adopted the planar space. Further studies are required to conduct a comprehensive investigation of transport networks by using network distance. Fourthly, the case study utilized only a few samples of activity-diary data. Although mobility big data has a huge number of samples, it lacks semantic contexts of activity-travel behaviors. Future studies are required to learn the semantic contexts of activity-travel behaviors for mobility big data by using emerging large language models and geospatial artificial intelligence. Last but not least, the proposed framework offers a computable and powerful approach for representing semantically enriched activity-travel behaviors. Further studies are required to apply this proposed framework to various time-geographical studies, such as accessibility, socio-spatial equity, and social segregation.

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Supplementary materials

S1. Semantic POI and semantic place entities defined in the physical space

This supplementary section introduces two types of semantic spatial entities, namely *SemPOI* and *SemPlace*, in the land-use systems of the physical space. As shown in Figure S1, a *POI* represents a point of interest (POI) in the physical space, such as a shopping center, school, and gym. Each POI, denoted by $SemPOI_i$, is expressed as a tuple:

$$SemPOI_i = \{\langle x_i, y_i \rangle, sm_SemPOI_i\} \quad (S1)$$

where $\langle x_i, y_i \rangle$ is its spatial location and sm_SemPOI_i is the set of semantic information. We further define $sm_SemPOI_i = \{Address, PoiType, OpenHour, CloseHour\}$. For example, the gym entity is represented by a tuple as $Gym_i = \{\langle 114.365, 30.537 \rangle, sm_Gym_i\}$. Using a similar representation technique, several sub-types of POI can be defined.

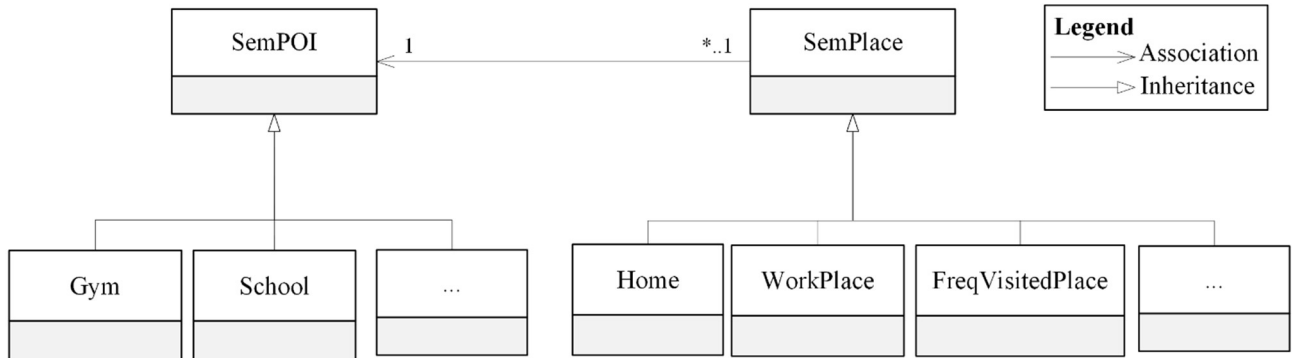


Figure S1. The concept model of *SemPOI* and *SemPlace* entities.

The semantic information of POI is kept the same for all individuals. We introduce the *SemPlace* entity to model the physical space with individual contexts, such as *Home*, *Workplace*, *Frequently Visited Places*, etc. A *semantic place* object, denoted by $SemPlace_j$, is expressed as a tuple:

$$SemPlace_j = \{SemPOI_i, sm_SemPlace_j\} \quad (S2)$$

where $SemPOI_i$ is the associated physical location, and $sm_SemPlace_j$ is the set of semantic information. For example, the Star Gym, $FreqVisitPlace_j^q$, is a frequently visited place of an individual q for sporting. Using a similar representation technique, several semantic places can be defined.

S2. Semantic cyber apps defined in cyberspace

This supplementary section introduces a semantic entity, namely *SemWebsite*, to represent an online application, e.g., Amazon, YouTube, Facebook, and Google. Each semantic object, denoted by $SemWebsite_i$, can be expressed as a tuple:

$$SemWebsite_i = \{sm_SemWebsite_i\} \quad (S3)$$

where $sm_SemWebsite_i$ is the set of semantic information and can be defined as $sm_SemWebsite_i = \{InternetAddress, ActivityType, OtherContextSet\}$

S3. Summary of semantic time-geographical entities

This section provides a summary of semantic time-geographical entities introduced in the proposed framework. Table S1 gives this summary with a comparison to entities defined in the classical notation system (Miller, 2005a). Along the line of the classical notation system of time geography, the proposed framework introduced four semantic time-geographical entities (i.e., semantic space-time path, station, prism, and lifeline) to represent activity-travel behaviors in the physical space. Several sub-entities are defined to incorporate semantic contexts at various levels of space-time granularities. Further, the proposed framework defines three semantic time-geographical entities (i.e., semantic website-time path, hybrid activity pattern, and hybrid project) to represent activity-travel behaviors in the hybrid cyber-physical space.

Table S1. Summary of semantic space-time entities introduced in the proposed notation system.

Classical notation system*	Proposed notation system		
Classical entities	Introduced entities	Sub-entities	Incorporated semantic contexts
Space-time path	Semantic space-time path (SemPath)	-	Path level (e.g., activity no.)
		SemActivity	Activity level (e.g., activity goal)
		SemTrip	Trip level (e.g., trip duration)
		SemSubtrip	Subtrip level (e.g., transport mode)
		SemSeg	Segment level (e.g., speed, heart rate)
Space-time station	Semantic space-time station (SemStation)	-	Station level (e.g., service type)
		SemSeg	Segment level (e.g., waiting time)
Space-time prism	Semantic space-time prism (SemPrism)	-	Prism level (e.g., activity goal)
		<i>SemSTCuboid</i>	Cuboid level (e.g., maximum duration)
		<i>SemSTCube</i>	Cube level (e.g., frequency of visits)
Space-time lifeline	Semantic space-time lifeline (SemLifeline)	-	Lifeline level (e.g., out-of-home time)
		SemPath	Path level (e.g., activity no.)
		SemPrism	Prism level (e.g., activity goal)
-	Semantic website-time path (<i>VirPath</i>)	-	Path level (e.g., activity no.)
		<i>VirActivity</i>	Activity level (e.g., activity goal)
-	Hybrid activity pattern (HybActivityPattern)	-	Activity pattern level (e.g., <i>HybEvent</i> No.)
		<i>HybEvent</i>	<i>HybEvent</i> level (e.g., isHybrid)
-	Hybrid project (HybProject)	-	Project level (e.g., activity goals)
		<i>HybEvent</i>	<i>HybEvent</i> level (e.g., isHybrid)

*Entities defined in Miller (2005a).

S4. Company entity defined in the social space

This supplementary section introduces a semantic company entity (*SemCompany*). A semantic company object, denoted by $SemCompany^c$, consists of multiple employees and has semantic information in terms of company attributes. It can be expressed as

$$SemCompany^c = \{EmpolyeeSet^c, sm_SemCompany^c\} \quad (S4)$$

where $EmpolyeeSet^c$ is the set for recording employees in the company; and $sm_SemCompany^c$ is the set for recording semantic information of the company. We set $sm_SemCompany^c = \{Address, EmployeeNum, CompanyType, OtherContextSet\}$. The $EmployeeNum$ attribute represents the number of employees in the company; the $CompanyType$ attribute represents the type of the company.

S5. Illustrative example of semantic space-time station

This supplementary section provides an illustrative example of a semantic space-time station entity. Figure 5 illustrates an example of a semantic space-time station, $SemStation^{KWT}$ for a famous restaurant, namely KWT. As shown, its $SemSubP^{KWT}$ element had three semantic line segments ($SemSeg_1^{KWT}$, $SemSeg_2^{KWT}$ and $SemSeg_3^{KWT}$) maintaining two semantic information, including opening status and waiting times. Then, the $OtherContextSet$ element of smt_SemSeg was set as $(opening\ status, waiting\ times)$. The $GeoSeg_{ij}$ element of all three segments was the vertical lines located at the restaurant. For $SemSeg_1^{KWT}$ object, its set of semantic information, $sm_SemSeg_1^{KWT} = \{0, (true, 0min)\}$. The other two segments, $SemSeg_2^{KWT}$ and $SemSeg_3^{KWT}$, had different semantic information as $sm_SemSeg_2^{KWT} = \{0, (true, 10min)\}$ and $sm_SemSeg_3^{KWT} = \{0, (true, 0min)\}$, respectively. The set of semantic information recorded in $sm_SemStation^{KWT} = \{KWT, null\}$, where KWT is the corresponding $SemPOI$. Figure 5 also illustrates another example of a semantic space-time station for a weather station, denoted by $SemStation^{Weat}$. The time-dependent weather condition information can be maintained in the $SemSegSet^{Weat}$ element of the weather station.

S6. An illustrative example of a semantic website time path entity

This supplementary section describes an illustrative example of the semantic website-time path entity. Figure S2 illustrates online activities in the 2D website-time space with a horizontal axis representing online applications (i.e., websites) and a vertical axis representing the time dimension. For example, there is online shopping, $VirActivity_1^{Lisa}$, conducted by Lisa from 7:20 to 7:50 through Amazon. We can record the set of semantic information as $sm_VirActivity_1^{Lisa} = \{Online\ Shopping, Amazon, 7:20, 7:50, 30min, null\}$. Accordingly, this online shopping activity could be represented by a vertical line at the Amazon location. The illustrative semantic portal-time path ($VirPath_1^{Lisa}$) consists of six online activities conducted by Lisa during a working day. Then, we have $VirActivitySet_1^{Lisa} = \{VirActivity_1^{Lisa}, ..., VirActivity_6^{Lisa}\}$, and $sm_VirPath_1^{Lisa} = \{6, 175min, null\}$.

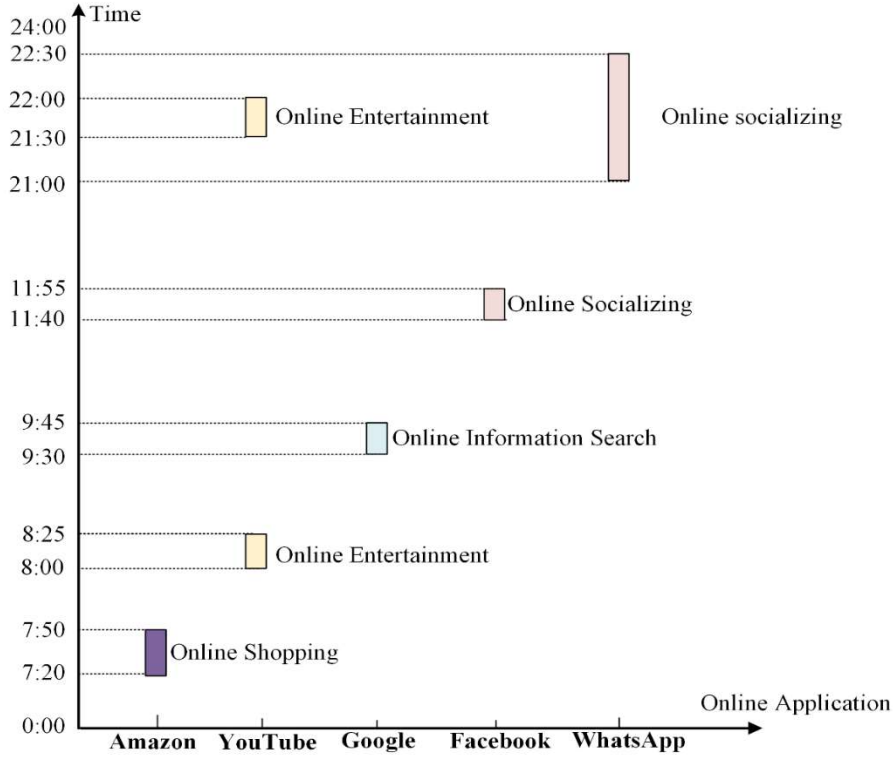


Figure S2. An illustrative example of a website-time path in the virtual space.

Compared to the semantic space-time path shown in Figure 5, this semantic website-time path has several unique characteristics. Firstly, the semantic website-time path is discontinuous. In the physical space, a semantic space-time path should be continuous in both space and time dimensions, given space-time constraints that the individual should be somewhere at any time, and a movement is required to overcome the distance between two locations (Shaw, 2023). These constraints can be relaxed in the semantic website-time path. The individual can be absent in the virtual space for any time episode, and no movement is required in the website-time space to switch to different online applications (or websites). Secondly, the semantic website-time path can have multiple online activities parallel in the time dimension. In the physical space, an individual has only one semantic space-time path during a typical day, because of the constraint that the individual is indivisible and can only be at one location at a time (Li et al., 2024; Ellegård, 2018). This constraint can also be relaxed in the semantic website-time path. For example, Lisa watches YouTube videos during 21:30-22:00 (i.e., $VirActivity_6^{Lisa}$); and conducts online socializing via WhatsApp (i.e., $VirActivity_5^{Lisa}$) during the same periods.

S7. Entity-relationship diagram

This supplementary section describes the ER (entity-relationship) diagram for implementing the proposed semantic time-geographic framework shown in Figure 2. As shown in Figure S3, every semantic time-geographic entity is transferred into a table in the relational database. Each table consists of three components: a primary key, attributes representing entity properties, and foreign keys. The primary key serves as a unique identifier for each semantic time-geographic entity, ensuring that every entity within the table can be uniquely identified. Attributes store semantic information about the entity, such as *Name*, *FamilyRole*, and *Occupation* attributes in the *SemIndividual* table. Foreign keys contain primary key values from other tables, establishing 1:1 or 1:N relationships between tables (i.e., semantic time-geographic entities). For instance, the *Home* attribute in *SemHousehold* can only be associated with one *SemPlaceID*, indicating the home location of the household; while *SemHousehold* can be associated with multiple *SemIndividualID*s, representing the composition of individuals within a household.

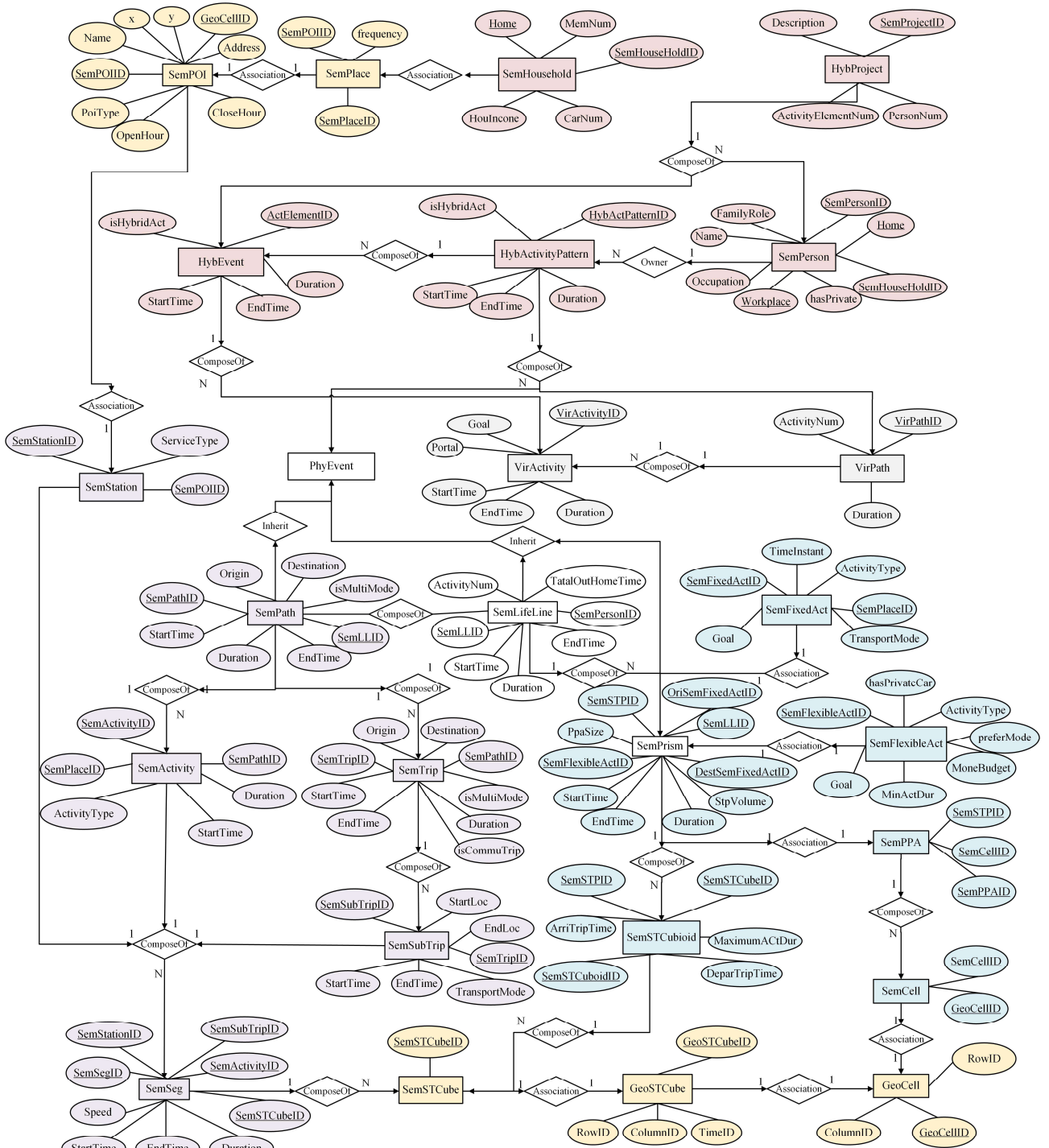


Figure S3. Entity-relationship diagram for semantic time-geographic entities in the relational database.

S8. User interface of the implemented framework

This supplementary section gives the user interface of the implemented framework using MySQL database and ArcGIS Pro software (see Figure S4).

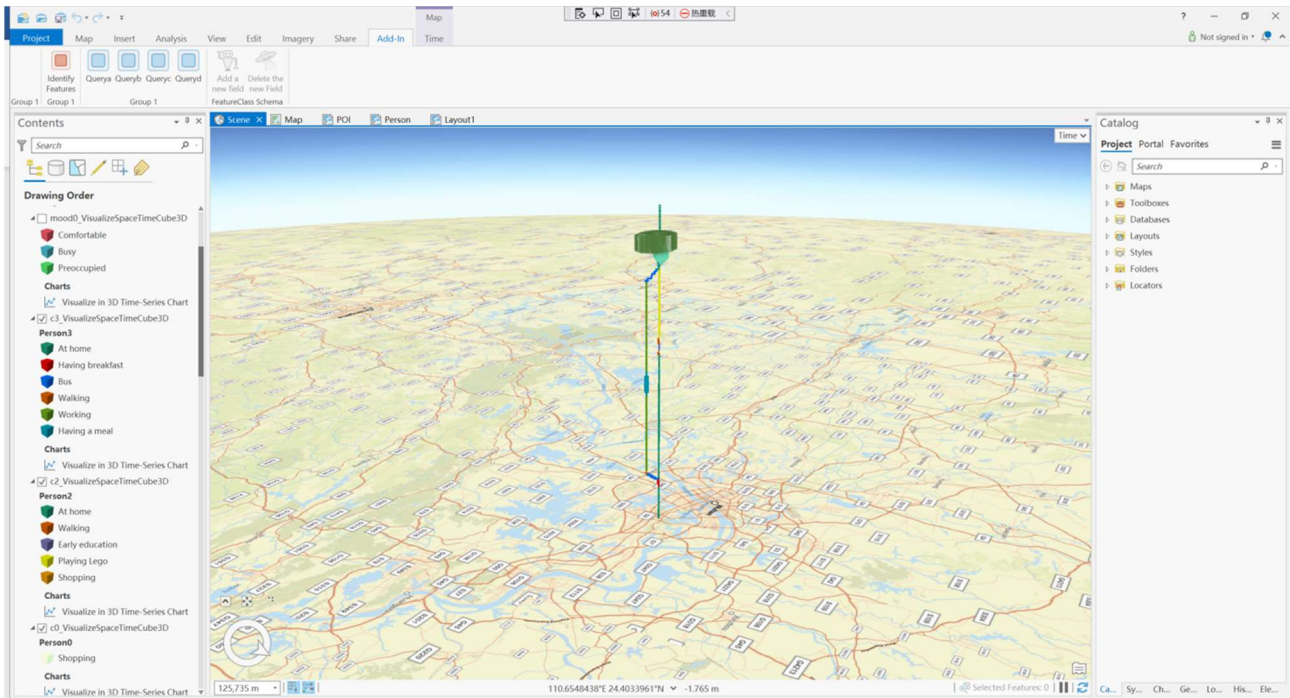


Figure S4. User interface of the implemented framework.

S9. Description of the first dataset in the case study

This supplementary section describes the activity diary data of 44 individuals used in the case study. Data were collected from a local website of Wuhan (<http://wh.bendibao.com>) between 20 July and 10 August 2021. Among these data, 14 individuals were collected from six families (i.e., with family information), while the other 30 individuals lacked their family information. Table S2 reports the collected information from six households, including the number of people, monthly income, and home location. Table S3 describes the working members of these six families. For the privacy issue, their real names were not used. In this dataset, everyone has their patterns of daily activities with detailed semantic information. Here are two examples:

The activity pattern of Nancy in Family #1 on 2 August 2021 is:

00:00-13:25 Staying at home

08:05-08:10 *Online socializing with her husband (added by authors for demonstration)*

09:00-12:00 *Teleworking at home (added by authors for demonstration)*

13:25-13:55 Taking the daughter to the Mengkana Picture Book Museum by walking

13:55-14:15 Taking the daughter to Early education in the Mengkana Picture Book Museum by walking

14:15-14:30 Taking the daughter to Beiqikechuang Robot Store by walking

14:30-14:45 Driving to Shenglong Electric Company

14:45-18:00 Working at Shenglong Electric Company

18:00-19:00 Driving home and purchasing food at a vegetable market on the way home

19:00-24:00 Staying at home

The activity pattern of John in Family #1 on 2 August 2021 is:

00:00-07:35 Staying at home

07:35-08:00 Walking to the Bus Station and Having breakfast on the way

08:00-08:27 Taking the Bus from Optics Valley Biological Park Bus Station

08:05-08:10 *Online socializing with his wife (added by authors for demonstration)*

08:10-08:25 *Online shopping (added by authors for demonstration)*

08:27-08:30 Walking to Zhenkunhang Industrial Supermarket

08:30-12:00 Working at Zhenkunhang Industrial Supermarket
12:00-13:00 Having lunch near Zhenkunhang Industrial Supermarket
13:00-17:00 Working at Zhenkunhang Industrial Supermarket
17:00-17:03 Walking to Optics Valley Biological Park Bus Station
17:03-18:45 Having dinner with friends and walking home
18:45-24:00 Staying at home

Table S2. Summary of the six households in the first dataset.

HID	Person_num	Income (RMB/ month)	Home_place
1	3	27000	Wanke Garden
2	2	-	Hongjing Community
3	2	8000	Guolingli Community
4	2	7500	Hongjing Community
5	2	-	Audit Bureau Family Court
6	3	-	Changqing Garden Fourth Community

Table S3. Summary of the 14 individuals with family information in the first dataset.

Name	HID	Role	Occupation	Work_place
John	1	Husband	Sales	Zhenkunhang Industrial Supermarket
Nancy	1	Wife	Electrical engineer	Shenglong Electric Company
Daisy	1	Daughter	-	-
Henry	2	Husband	Worker	China Construction Third Bureau
Nina	2	Wife	Worker	Xiangxiewan construction site
Adam	3	Husband	Worker	China Construction Third Bureau
Annalee	3	Wife	Worker	China Construction Third Bureau
Carmen	4	Husband	Worker	China Construction Third Bureau
Carry	4	Wife	Housewife	At-home
Brent	5	Father	-	-
Bruce	5	Son	-	-
Chenney	6	Mother	Peddler	Central Harbor Market
Ben	6	Son	-	-
Cecil	6	Daughter	-	-

S10. Description of the procedure to simulate the data used in the second dataset

This supplementary section describes the procedure to simulate the data used in the second dataset of the case study. The used procedure consists of four steps. The first step is to generate the household structure and home location information. The household structure is generated by randomly selecting one household from the six households in the first dataset to obtain the number of household members and their roles. The home location is generated by randomly selecting a community in Wuhan City. The second step is to generate occupation and workplace location information for each household member. For a child, we simply assigned the child's school location as the nearest school to the home location. For household heads, we generated the job opportunity set by searching nearby POIs for opportunities within a 20 km buffer zone from their home location. Then, we randomly selected a POI within the job opportunity set as his/her workplace and assigned the occupation information according to the corresponding type of POI. The third step is to generate activity program information by randomly selecting one from the activity program of 44 individuals collected. The last step is to generate a semantic lifeline for each individual. According to the generated activity program information, we first assigned fixed activities (e.g., working, taking children to school, etc.) and then assigned flexible activities and/or trips between any two subsequent fixed activities. We considered three scenarios according to the number of flexible activities between two subsequent fixed activities.

If there is no flexible activity between them, we use the transit shortest path algorithm to plan a multi-mode journey (i.e., generating a multi-mode trip). If there is only one flexible activity, we generate a semantic space-time prism. If there is more than one flexible activity, we simply merge them into a flexible activity and generate a semantic space-time prism.

S11. Detailed descriptions of four queries tested in the case study

This supplementary section describes detailed information about four representative queries tested in the Section of Query Results. The second dataset (simulated dataset) was used for query performance testing. Table S4 summarizes the sizes of 29 tables in this dataset.

Table S4. The sizes of 27 tables were used for the framework implementation.

Table Name	Size (10 ³)	Table Name	Size (10 ³)
SemHousehold	5.0	SemPrism	15.48
SemIndividual	10.97	SemSTCuboid	2392.67
SemIndividual_Individual_Cont	10.9	SemSTCuboid_STCube_Cont	29293.14
SemLiefLine	10.24	SemPPA	15.48
SemPath	23.46	SemPPA_Cell_Cont	2392.67
SemActivity	31.14	SemCell	2392.67
SemActivity_Seg_Cont	31.14	GeoCell	6.7
SemTrip	12.66	GeoSTCube	1942.95
SemSubTrip	14.19	GeoCell_Cell_Cont	52.8
SemSubTrip_Seg_Cont	14.19	GeoSTCube_Cell_Cont	1942.95
SemSeg	45.32	GeoSTCube_STCube_SpatNeib	15260.36
SemSeg_STCube_Cont	3463.11	GeoSTCube_STCube_TempNeib	3872.45
SemSTCube	32756.25	SemPlace	5.0
HybActivityPattern	5.0	SemPOI	126.22
HybEvent	60.81		

Using a relational database, we can employ structured query language (SQL) to implement various queries on semantic time geographical entities. We implemented four types of queries in this case study. The first type is person-based queries, which search for semantic time geographical entities of specific persons. We employed Query (a) for demonstration: Where is John, and what is he doing at 12:00? Given the name attribute of the Individual (i.e., John), this illustrative search is to retrieve the semantic time geographical entities satisfying the time constraint (i.e., 12:00). It involves the joint queries on sixteen tables (i.e., *SemIndividual*, *SemLiefLine*, *SemPath*, *SemActivity*, *SemActivity_Seg_Cont*, *SemSeg*, *SemSeg_STCube_Cont*, *SemSTCube*, *GeoSTCube*, *SemTrip*, *SemSubTrip*, *SemSubTrip_Seg_Cont*, *SemPrism*, *SemSTCuboid*, *SemSTCuboid_STCube_Cont*, and *GeoCell*).

The second type is semantic-based queries, which retrieve entities based on their semantic attributes. We used Query (b) for illustration: Who are the friends of individuals whose monthly income exceeds 10,000 RMB? It involves the joint queries on two tables (including *SemIndividual* and *SemIndividual_Individual_Cont*).

The third type is relationship-based queries, which search for information according to semantic space-time relationships. We employed Query (c) for illustration: Who is likely to have lunch with John? Given the name attribute of the person, this search firstly retrieves the person's semantic space-time entities associated with his lunch activity and then retrieves individuals whose semantic time geographical entities have the semantic space-time intersection relationship with that of John. It involves the joint queries on fifteen tables (i.e., *SemIndividual*, *SemLiefLine*, *SemPath*, *SemActivity*,

SemActivity_Seg_Cont, *SemSeg*, *SemSeg_STCube_Cont*, *SemSTCube*, *GeoSTCube*, *SemTrip*, *SemSubTrip*, *SemSubTrip_Seg_Cont*, *SemPrism*, *SemSTCuboid*, and *SemSTCuboid_STCube_Cont*).

The fourth type is space-time-based queries, which search entities based on input space-time regions. We utilized Query (c) for illustration: Who visited the region within 2km around the Shopping Center “QunGuang” from 18:00 to 19:00? Given the name attribute of the Shopping Center, this search firstly retrieves the GeoCell of the shopping center, constructs its 2 km buffer zones, determines the space-time regions (i.e., lifting buffer zones with the time dimension from 18:00 to 19:00), and finally searches the semantic time geographical entities within the determined space-time regions. This search involves joint queries on sixteen tables (i.e., *SemPOI*, *SemIndividual*, *SemLiefLine*, *SemPath*, *SemActivity*, *SemActivity_Seg_Cont*, *SemSeg*, *SemSeg_STCube_Cont*, *SemSTCube*, *GeoSTCube*, *SemTrip*, *SemSubTrip*, *SemSubTrip_Seg_Cont*, *SemSTP*, *SemSTCuboid*, and *SemSTCuboid_STCube_Cont*).

The computational performance of these four types of queries is reported in Section of Query Results. The user interfaces for these four types of queries are given below.

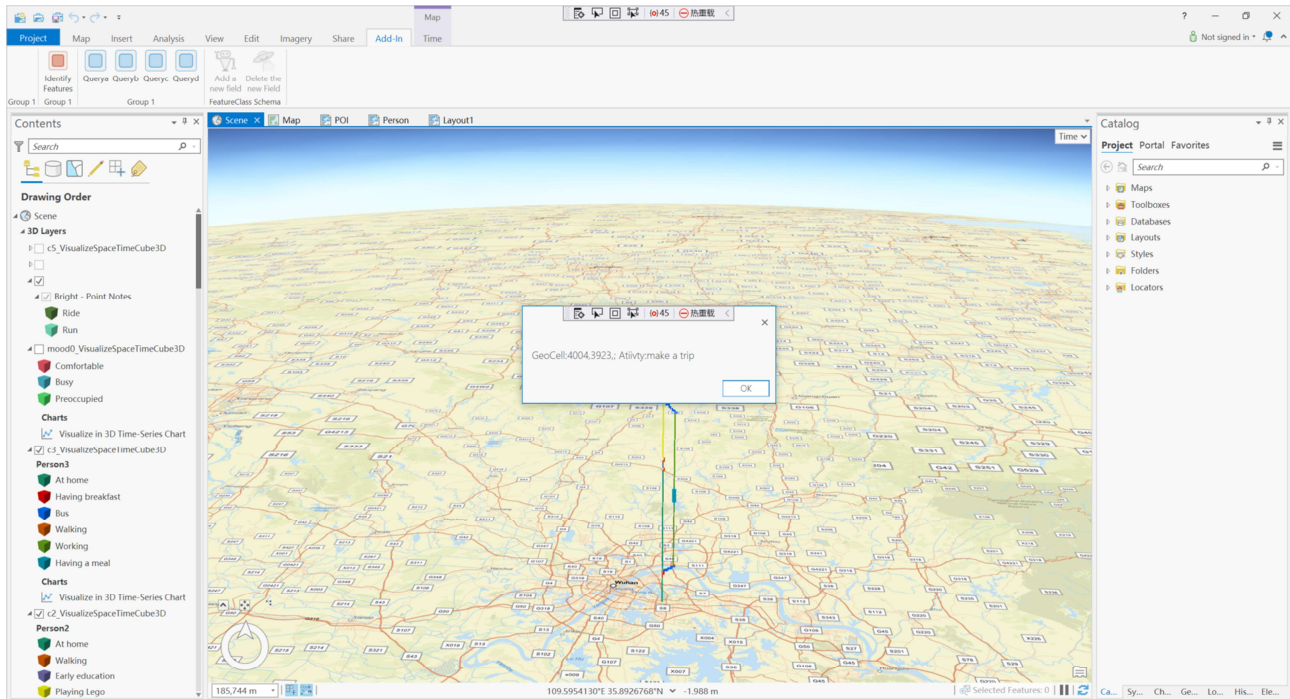


Figure S5. The user interface and result of Query (a).

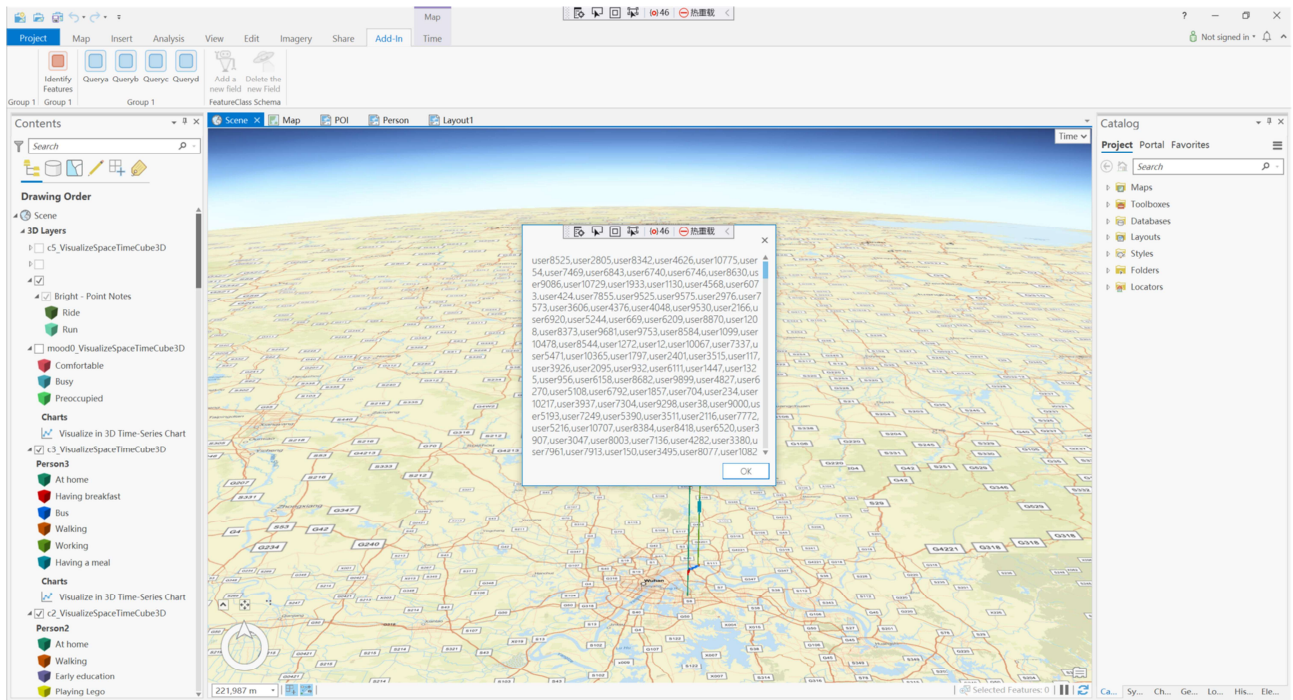


Figure S6. The user interface for the illustrative Query (b).

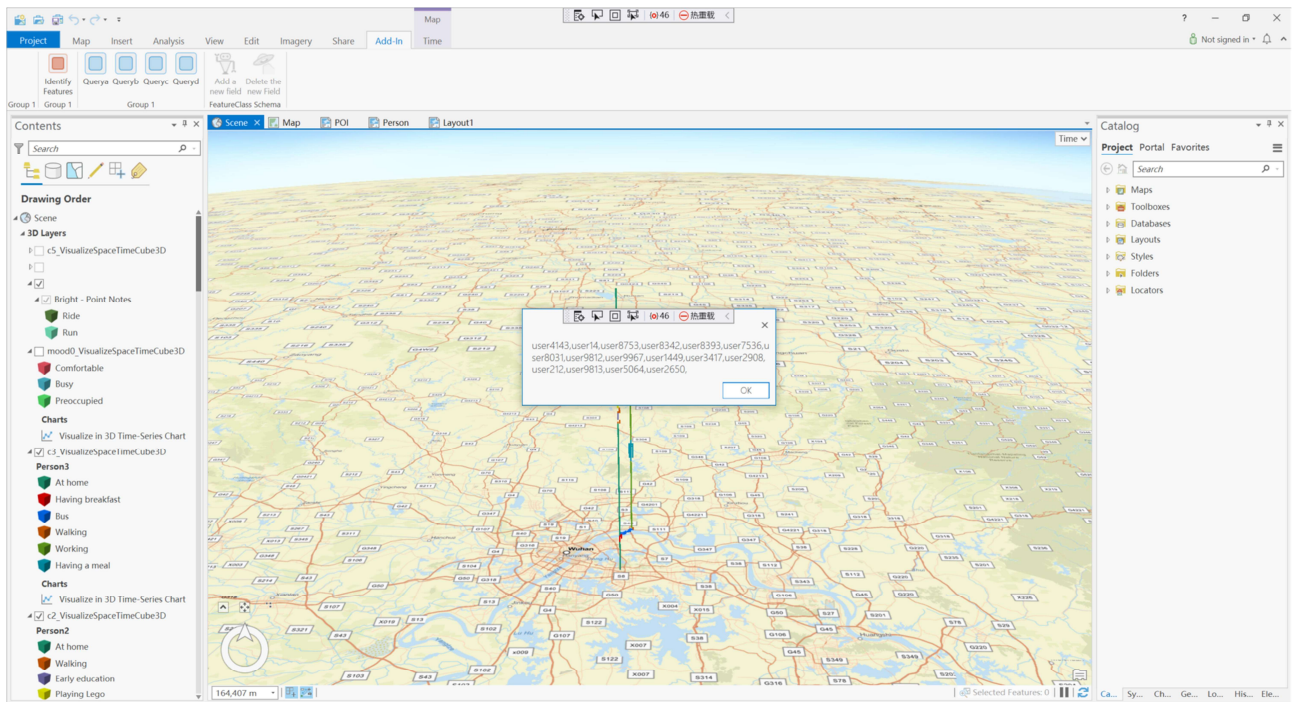


Figure S7. The user interface and result of Query (c).

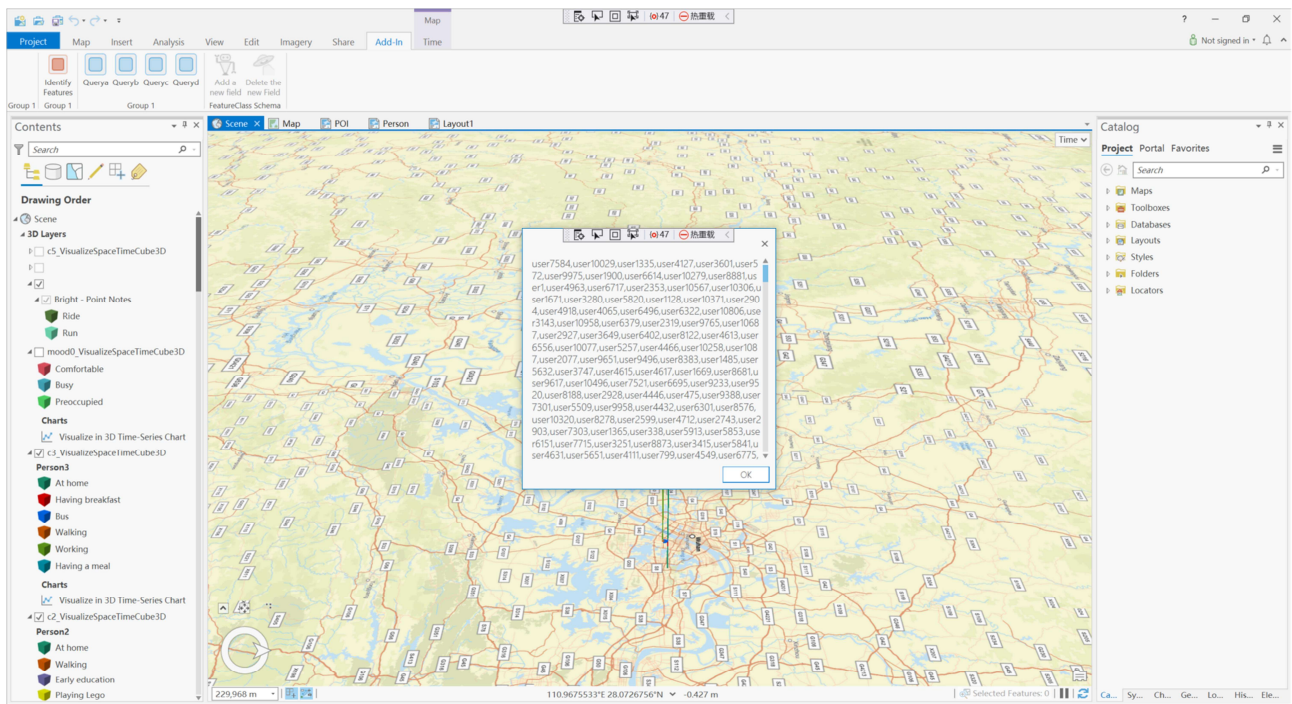


Figure S8. The user interface and result of Query (d).